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Novel item selection strategies for cognitive diagnostic computerized adaptive testing: A heuristic search framework

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Abstract

The computerized adaptive form of cognitive diagnostic testing, CD-CAT, has gained increasing attention in the domain of personalized measurements for its ability to categorize individual mastery status of fine-grained attributes more accurately and efficiently through administering items tailored to one's ability progressively. How to select the next item based on previous response(s) is crucial for the success of CD-CAT. Previous item selection strategies for CD-CAT have often followed a greedy or semi-greedy approach, which makes it difficult to strike a balance between diagnostic performance and item bank utilization. To address this issue, this study takes a graph perspective and transforms the item selection problem in CD-CAT into a path-searching problem, in which paths refer to possible test construction and nodes refer to individual items. A heuristic function is defined to predict the prospect of a path, indicating how well the corresponding test can diagnose the current examinee. Two search mechanisms with different biases towards item exposure control are proposed to approximate the optimal path with the best prospect. The first unused item on the resulting path is selected as the next item. The above components compose a novel CD-CAT item selection framework based on heuristic search. Simulation studies are conducted under a variety of conditions regarding bank designs, bank-quality conditions, and testing scenarios. The results are compared with different types of classic item selection strategies in CD-CAT, showing that the proposed framework can enhance bank utilization at a smaller cost of diagnostic performance.

Keywords Cognitive diagnostic testing · Computerized adaptive testing · Heuristic search

Introduction

Cognitive diagnosis models (CDMs) are a class of psychometric models that measure the statuses of examinees on fine-grained attributes of a subject (Jiang & Carter, 2018; Leighton & Gierl, 2007; Zhang & Chang, 2019). Compared with conventional psychometric models that assess examinees from a unidimensional perspective (e.g., item response theory and classic test theory), CDMs provide more detailed information of examinees, and thus are of great value in promoting personalized assessment. Computerized adaptive testing (CAT) is a computer-based measurement (Meijer & Nering, 1999; Weiss & Kingsbury, 1984). Different from traditional paper-and-pencil (P&P) testing that preassembles the entire test before administration, CAT selects each item in real-time according to the estimated ability of an examinee based on the response(s) to previous item(s) and thus tailors the test to the examinee's personal status. Compared to P&P, CAT is able to achieve higher measurement precision and efficiency (Chang, 2015; Chang & Ying, 1999; Chen & Cook, 2009; Gibbons et al., 2012; Wang et al., 2011a), thus raise broad attention in vast context, including education (Mizumoto et al., 2017; Tseng, 2016), psychiatry (Gibbons & de Gury, 2019; Gibbons et al., 2020; Moore et al., 2015), and clinic (Boody et al., 2018; Gausden et al., 2018).

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Cognitive diagnostic computerized adaptive testing (CD-CAT) is CAT developed based on CDMs. By combining the fine-grained diagnostic capability of CDMs and the adaptive testing ability of CAT, CD-CAT has received increasing interest from the fields of educational measurement and psychometrics. The item selection strategy, which decides how to select the next item for an examinee, is one of the most critical components of CD-CAT (Yu et al., 2019). Early efforts in this regard focus on increasing the diagnostic efficiency and accuracy. To do so, a variety of criteria are designed to evaluate the utility of an item in promoting fast and precise diagnosis of the current examinee based on the response(s) to previous item(s). Correspondingly, various item selection strategies have been developed to greedily determine the next item as the optimal one regarding the specified criterion. In the literature, the most influential strategies include the Kullback–Leibler information (KL) (Xu et al., 2003), Shannon entropy (SHE) (Tatsuoka & Ferguson, 2003), posterior-weighted KL (PWKL) (Cheng, 2009), mutual information (MI) (Wang, 2013), and generalized model discrimination index (GDI) (Kaplan et al., 2015).

Nevertheless, although these strategies offer high measurement accuracy, they bring along the problem of disproportionate item bank utilization. That is, some items in the bank are used much more frequently than others, while some items are rarely used. Item overexposure may yield unexpected familiarity, making the related items easier to guess for subsequent examinees. Therefore, excessive item overexposure will jeopardize test security and impair diagnostic validity. This issue has become increasingly serious in recent years as cognitive diagnostic tests exhibit more and more potential for high-stakes assessments that have higher requirements for both security and accuracy (Ravand, 2016; Hemati & Baghaei, 2020). On the other hand, the underutilization of items causes a waste of bank capacity. CDM items require more parameters (e.g., Q-vector) than items in traditional test models such as item response theory (IRT). The development and calibration of CDM items thus pose a higher demand on human efforts, time investments, and financial resources (Li & Suen, 2013; Lim & Drasgow, 2017; Wang et al., 2011b), which makes it particularly important to minimize the waste of bank capacity in CD-CAT. Therefore, it is crucial for item selection strategies to have the ability to balance item bank utilization and avoid extensive item overexposure and underutilization.

A number of item selection strategies that consider bank utilization have been developed in recent years. Wang et al. (2011b) presented two methods based on PWKL, namely restrictive progressive PWKL (RP_PWKL) and restrictive threshold PWKL (RT_PWKL). RP_PWKL imposes a compulsory exposure restriction on the item selection criterion in a modified PWKL method, while the RT_PWKL method conducts random selection in an item set spanning the near-optimum interval of the PWKL index and uses the exposure degree to regulate the size of the interval. Zheng and Wang (2017) considered the CD-CAT problem as a dynamic searching problem and proposed a binary searching method (DBS) that randomly selects the next item from an item set in which the attribute measuring patterns can evenly split the posterior probabilities of all possible attribute mastery patterns into half. However, despite some success in controlling item exposure, the above strategies generally suffer from significant degeneration in terms of diagnostic efficiency and accuracy.

The current study aims at developing novel item selection strategies for CD-CAT to balance bank utilization without seriously impairing diagnostic performance. The existing item selection strategies, either focusing on diagnostic capability or considering item exposure control, are mostly greedy or semi-greedy algorithms, that is, always choosing the currently (near-)optimal item regarding certain criterion. Such greediness limits the diversity of selected items, which may be the primary cause of unbalanced bank utilization. Incorporating random components can increase selection diversity, but at the risk of lowering the diagnostic performance for involving unsuitable or even noisy items. To address the above issue, this study views the item bank as a graph and transforms the item selection problem in CD-CAT into a path-searching problem on the graph, in which paths refer to constructing tests and nodes refer to items. The goal of the path-searching problem is to find the optimal path corresponding to a test that is predicted to best diagnose the current examinee based on the existing response data. Once found, the first unvisited node on the optimal path, i.e., the first unused item, is selected and administered to the examinee. After the examinee responds, the path-searching problem restarts based on the updated response data to determine the next item. A tree-based heuristic search mechanism is designed to solve the above path-searching problem. The mechanism defines a heuristic function that can estimate the diagnostic prospect of a path and introduces two heuristic search strategies that generate the searching trees with different biases. The above path-searching idea could be connected to the idea of looking ahead in IRT-based CAT (van der Linden, 1998; van der Linden & Glas, 2000a). That is, they both consider two or more items ahead for selecting the next item. However, the path-searching idea builds each path by adding eligible items one by one. This incremental learning manner facilitates the control of extra computational cost introduced by the consideration of additional items (e.g., by pruning or sampling). In contrast, the looking-ahead idea considers item(s) ahead in a manner of enumerating all possible item combinations. The computational complexity thus grows exponentially with the bank size, making it only favorable for small item banks (van der Linden, 1998; van der Linden & Glas, 2000a). Moreover, the idea of looking

ahead is conceived in the context of IRT (Jiao et al., 2019; Li et al., 2019; van der Linden, 1998, 2008; van der Linden & Glas, 2000a) and cannot be easily adapted to CD-CAT due to their essential difference (Sorrel et al., 2020). Our path-searching idea is the initial attempt for considering more items ahead for item selection in CD-CAT. Simulation studies are conducted to validate the effectiveness of the proposed item selection framework under a variety of conditions. The results are compared with the classic item selection strategies in the CD-CAT literature to further illustrate the advantage of the proposed framework.

The rest of this paper is organized as follows. "Background" section presents a brief review of the CD-CAT problem and the classic item selection strategies. Detailed implementation of the proposed strategies is introduced in "Heuristic-search-based item selection strategies" section. Simulation design and results in comparison with the classic item selection strategies are reported in "Simulation study" section. "Discussion and conclusion" section concludes this study and highlights several future directions.

Background

This section first introduces CDMs and their computerized adaptive form. Then, we briefly review the classic item selection strategies in CD-CAT.

Cognitive diagnostic tests and their computerized adaptive form

CDMs define the item response functions (IRFs) based on the attributes required by items and the attribute mastery patterns (AMP) of examinees. Take the reduced reparameterized unified model (r-RUM) (Hartz, 2002) as an example. The r-RUM adopts a dichotomous assumption, meaning that the attribute mastery status, attribute requirements per item, and item responses are all binarized. Suppose that there are I items measuring K attributes. For each item i $(i = 1, 2, \ldots, I)$, let $\boldsymbol{q}_i = (q_{i1}, q_{i2}, \ldots, q_{iK})$ represent the *Q-vector* that specifies the attribute requirements and σ_i denote the response. Given an AMP $\boldsymbol{\alpha} = (\alpha_1, \alpha_2, \ldots, \alpha_K)$, the IRFs are

$$P\left(\sigma_i = 1 \middle| \boldsymbol{\alpha}\right) = \widehat{\pi}_i \prod_{k=1}^{K} \widehat{r}_{ik}^{(1-\alpha_{ik})q_{ik}} \tag{1}$$

and $$P\left(\sigma_i = 0 \middle| \boldsymbol{\alpha}\right) = 1 - P\left(\sigma_i = 1 \middle| \boldsymbol{\alpha}\right), \tag{2}$$

where $\widehat{\pi}_i \in [0, 1]$ is the baseline parameter denoting the correct response probability given that all the required attributes are mastered and $\widehat{r}_{ik} \in [0, 1)$ is the penalty parameter denoting the penalizing ratio on the correct response probability if a required attribute k is not mastered. Given a cognitive diagnostic test (CDT) where the attribute requirements of the used items are already specified with a *Q-matrix*, the latent AMP of an examinee can be estimated by fitting their responses according to the IRFs. Note that the approach proposed in the current study is independent of CDMs. Yet we will continue to take r-RUM as an example in the following illustration for facilitating presentation.

Unlike the preassembled CDT that administrates the same set of items to all examinees, CD-CAT offers a customized test to each examinee through online adaptive search. That is, it assigns one item to the examinee at a time. Once the examinee responds, CD-CAT then selects the next item based on the previous response(s) and repeats this operation until the termination criterion is satisfied, e.g., the AMP estimation converges, or the preset test length is reached. By doing so, CD-CAT can shorten the test length and provide higher diagnostic efficiency than conventional CDT.

Classic item selection strategies in CD-CAT

This subsection briefly reviews four classic item selection strategies in the CD-CAT literature, including two of the most popular strategies that focus on diagnostic performance and two notable strategies that embed item exposure control. These strategies are also involved for comparison in the subsequent simulation studies. To facilitate descriptions, we would like to define several notations that will be used in the following text. Given an I-item bank Ω, $\boldsymbol{x}$ denotes a complete adaptive test built from Ω and $\overline{\boldsymbol{x}} = [\overline{x}_1, \overline{x}_2, \ldots, \overline{x}_m]$ denotes its incomplete form with m items, i.e., m items have been administrated. $\overline{\boldsymbol{u}} = [\overline{u}_1, \overline{u}_2, \ldots, \overline{u}_m]$ represents the responses to $\overline{\boldsymbol{x}}$. Intrinsically, CD-CAT requires the next item to be selected from the unused set of Ω, i.e., $\{i \in \Omega \wedge i \notin \overline{\boldsymbol{x}}\}$. A series of addition operators are re-defined as follows:

- vector + [scalar]: $\boldsymbol{v} + s$ represents appending scalar s to vector $\boldsymbol{v}$.
- vector + vector: $\boldsymbol{v} + \boldsymbol{v}'$ represents appending all the elements in vector $\boldsymbol{v}'$ to vector $\boldsymbol{v}$.
- matrix + [vector]: $\mathbf{M} + \boldsymbol{v}$ represents appending vector $\boldsymbol{v}$ to matrix $\mathbf{M}$ as a new row.

Shannon entropy (SHE)

The Shannon entropy is an uncertainty measure for the probability distribution of a random variable (Shannon, 1948). Given a random variable with Γ possible values and its probability distribution $\boldsymbol{\pi} = [\pi(1), \pi(2), \ldots, \pi(\Gamma)]$, the Shannon entropy is defined by

$$Sh(\boldsymbol{\pi}) = -\sum_{\gamma=1}^{\Gamma} \pi(\gamma) \log[\pi(\gamma)]. \tag{3}$$

Tatsuoka and Ferguson (2003) introduced the Shannon entropy into the context of CD-CAT. Given a constructing test $\bar{\boldsymbol{x}}$ with m items, the interim posterior probability of AMP $\boldsymbol{\alpha}$ is

$$\pi_m(\boldsymbol{\alpha}) \propto \pi_0(\boldsymbol{\alpha}) \prod_{i=1}^{m} P\left(\sigma_{\bar{x}_i} = 1 \middle| \boldsymbol{\alpha}\right)^{\bar{u}_i} P\left(\sigma_{\bar{x}_i} = 0 \middle| \boldsymbol{\alpha}\right)^{1-\bar{u}_i}, \tag{4}$$

where $\pi_0(\boldsymbol{\alpha})$ is the prior probability of $\boldsymbol{\alpha}$. Then the Shannon entropy for the distribution of AMP interim posterior probability can be calculated by

$$Sh\left(\boldsymbol{\pi}_m\right) = -\sum\nolimits_{\boldsymbol{\alpha}} \pi_m(\boldsymbol{\alpha}) \log[\pi_m(\boldsymbol{\alpha})]. \tag{5}$$

A larger $Sh(\boldsymbol{\pi}_m)$ indicates higher uncertainty in the diagnostic result. Therefore, aiming to reduce diagnostic uncertainty, the criterion of the SHE strategy for selecting the next item is to minimize the expected Shannon entropy. More specifically, the expected Shannon entropy for an eligible item i can be computed by

$$SHE\left(\boldsymbol{\pi}_{m+1} \middle| i\right) = \sum\nolimits_{\sigma \in \{0,1\}} [Sh\left(\boldsymbol{\pi}_{m+1} \middle| \bar{\boldsymbol{u}} + [\sigma]\right) P(\sigma_i = \sigma | \boldsymbol{\pi}_m)]. \tag{6}$$

After calculating the expected Shannon entropy for all the eligible items in the bank, the one with the minimum value is selected as the $(m+1)^{\text{th}}$ item.

Generalized model discrimination index (GDI)

The generalized model discrimination index (GDI) introduces a reduced attribute vector $\acute{\boldsymbol{\alpha}}$ as a replacement for each original AMP $\boldsymbol{\alpha}$ when evaluating a candidate item (Kaplan et al., 2015). Unlike the original AMP that contains all attributes, the reduced attribute vector only involves the attribute(s) measured by the item. Based on the reduced attribute vectors of eligible item i, the GDI method quantifies the weighted variance in the correct response probabilities as

$$GDI_i = \sum\nolimits_{\acute{\boldsymbol{\alpha}}} \pi_m(\acute{\boldsymbol{\alpha}}) \left[P\left(\sigma_i = 1 \middle| \acute{\boldsymbol{\alpha}}\right) - \bar{P}_i\right]^2, \tag{7}$$

where $\pi_m(\acute{\boldsymbol{\alpha}})$ denotes the interim posterior probability of $\acute{\boldsymbol{\alpha}}$ that is estimated from the responses $\bar{\boldsymbol{u}}$ to the constructing test $\bar{\boldsymbol{x}}$ and $\bar{P}_i$ is the expected probability of correctly responding item i, i.e.,

$$\bar{P}_i = \sum\nolimits_{\acute{\boldsymbol{\alpha}}} \pi_m(\acute{\boldsymbol{\alpha}}) P\left(\sigma_i = 1 \middle| \acute{\boldsymbol{\alpha}}\right). \tag{8}$$

Kaplan et al. (2015) proposed that GDI indicates the capacity of an item to distinguish between AMPs regarding solely the attributes measured by the item. The GDI method thus selects the candidate item with the maximum GDI as the next item in CD-CAT.

Restrictive progressive posterior-weighted Kullback–Leibler information (RP_PWKL)

The Kullback–Leibler (KL) information, also known as the KL divergence, measures the distance between two probability distributions (Kullback & Leibler, 1951). Given two probability density functions $\boldsymbol{\pi}$ and $\boldsymbol{\varpi}$, the KL information is defined as

$$K(\boldsymbol{\pi}, \boldsymbol{\varpi}) = \int \log\left[\frac{\boldsymbol{\pi}(y)}{\boldsymbol{\varpi}(y)}\right] \boldsymbol{\pi}(y) \mathrm{d}y. \tag{9}$$

When the KL information is applied to CD-CAT, the next item is selected as the one that maximizes the total difference between the response probability distributions of the AMP estimated from the existing responses and the other AMPs (Xu et al., 2003). Given an item i, the KL information between the estimated AMP $\hat{\boldsymbol{\alpha}}$ and a competing AMP $\boldsymbol{\alpha}$ is calculated as

$$K_i\left(\hat{\boldsymbol{\alpha}}, \boldsymbol{\alpha}\right) = \sum\nolimits_{\sigma \in \{0,1\}} \log\left[\frac{P(\sigma_i = \sigma | \hat{\boldsymbol{\alpha}})}{P(\sigma_i = \sigma | \boldsymbol{\alpha})}\right] P(\sigma_i = \sigma | \hat{\boldsymbol{\alpha}}). \tag{10}$$

Hence, the overall KL information for item i is computed as

$$KL_i\left(\hat{\boldsymbol{\alpha}}\right) = \sum\nolimits_{\boldsymbol{\alpha}} K_i\left(\hat{\boldsymbol{\alpha}}, \boldsymbol{\alpha}\right). \tag{11}$$

However, the KL method implicitly assumes that all AMPs are equally likely to be the true AMP in the entire testing process. Cheng (2009) suggested using the interim posterior probabilities of AMPs to weight their KL information with the estimated AMP. The posterior-weighted KL (PWKL) information of item i would then be

$$PWKL_i\left(\hat{\boldsymbol{\alpha}}\right) = \sum\nolimits_{\boldsymbol{\alpha}} K_i\left(\hat{\boldsymbol{\alpha}}, \boldsymbol{\alpha}\right) \pi_m(\boldsymbol{\alpha}). \tag{12}$$

Wang et al. (2011b) later proposed a PWKL variant, namely P_PWKL, which progressively controls item exposure by adding a stochastic component to the PWKL information:

$$P_PWKL_i = \left(1 - \frac{m}{N}\right)\delta + PWKL_i \times \rho \frac{m}{N}, \tag{13}$$

where m refers to the number of items in the constructing test $\bar{\boldsymbol{x}}$, N is the predefined test length, δ is a random component subject to the uniform distribution $U(0, \mathcal{H}^*)$ with $\mathcal{H}^* = \max_i PWKL_i$, and ρ is a positive parameter that modifies the relative weight of PWKL. By doing so, the selection is prevented from always choosing the item with the highest PWKL information, which helps balance item exposure.

To further suppress item overexposure, another compulsory restriction can be imposed on P_PWKL:

$$RP_PWKL_i = \left(1 - \frac{\varrho_i}{o}\right) P_PWKL_i, \tag{14}$$

where ϱ_i refers to the current exposure rate of item i and o is the predefined threshold for item overexposure. By doing so, RP_PWKL can clamp the item exposure rate to o.

Dynamic binary searching (DBS)

Zheng and Wang (2017) exploited the thought of binary searching and proposed the DBS method, which favors the items whose Q-vector can evenly divide the interim posterior probabilities of AMPs in half. In DBS, a separation index $S_{i,\boldsymbol{\alpha}}$ is first defined to describe whether the AMP $\boldsymbol{\alpha}$ completely covers the attributes required by item i:

$$S_{i,\boldsymbol{\alpha}} = \prod_{k\in\Delta} \alpha_k, \Delta = \left\{k : q_{ik} = 1\right\}. \tag{15}$$

If $\boldsymbol{\alpha}$ possesses all the required attributes of item i, $S_{i,\boldsymbol{\alpha}} = 1$; otherwise, $S_{i,\boldsymbol{\alpha}} = 0$. Based on the separation indices, the binary searching index for item i is defined by

$$\mathcal{B}_i = \left|\sum\nolimits_{\boldsymbol{\alpha}\in\Delta} \pi_m(\boldsymbol{\alpha}) - 0.5\right|, \Delta = \{\boldsymbol{\alpha} : S_{i,\boldsymbol{\alpha}} = 1\}, \tag{16}$$

The items with $\mathcal{B}_i$ closer to zero are thus preferable. It is obvious that the items sharing the same Q-vector will get equal values for the binary searching index. The DBS method then randomly selects an item within those having the minimum binary searching index. By embedding such randomness, the DBS method is able to relieve item overexposure as it avoids repeatedly selecting the same item that has the best index value.

Heuristic-search-based item selection strategies

The current study adopts a graph perspective and regards the assembly of CD-CAT as a path-searching procedure. More specifically, the items in the given bank are graph nodes and the constructing test with one or more administered items is an incomplete path. The goal of path searching is to decide the next node (item) that leads to a complete path (test) that best diagnoses the current examinee. Define the search state as the distribution of AMP interim posterior probabilities, which is estimated based on the examinee's response to the constructing test. By the meaning of CAT, it is clear that the search state changes every time a new item is selected and answered, which may also cause the optimal path to switch, making path searching a dynamic problem.

To address this problem, we propose a tree-based heuristic search mechanism with the searching tree rooted from the currently constructing test. Each branch in the tree is a sequence of nodes denoting items that would be used in the test. We then define a heuristic function to predict the prospect of a branch according to the diagnostic performance if the corresponding items are added to the test. The number of branches allowed to be expanded from one node is the search width, the number of nodes allowed on each branch is the search depth, and the ordinal position in a branch is named the horizon. The search mechanism builds a tree according to the preset search width and depth, but only the node (item) at the first horizon of the most promising branch would be added to the test and administered to the examinee. After the examinee answers the new item, the search state is updated and the above tree-based search mechanism is repeated for selecting the next item until the whole test is completed. A flowchart summarizing the general process of item selection procedure is illustrated in Fig. 1.

In this section, we first define the heuristic function for evaluating a tree branch. Then we introduce two different

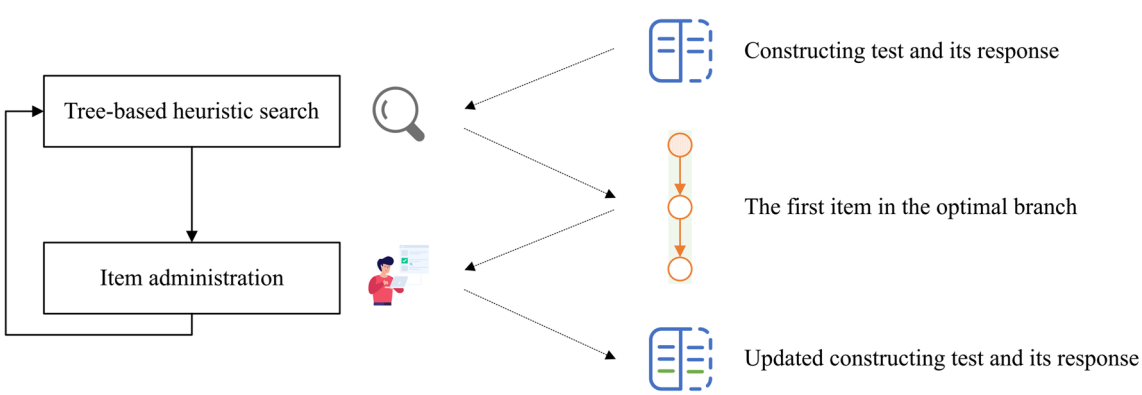

Fig. 1 Flowchart of the general item selection procedure

strategies to realize the tree-based search mechanism. Finally, we present the other components necessary to complete the proposed item selection strategies, including the constraint handling scheme and the first item selection scheme.

Definition of the heuristic function

We define a heuristic function based on the Bayesian estimation of examinees' AMPs to evaluate the prospect of a branch in the tree that is generated during the path-searching procedure. The heuristic function is evaluated through the following three steps:

Step 1 - Estimation of the present search state: Given a constructing test $\bar{\boldsymbol{x}}$ with m items, the posterior probability for an examinee that has a response vector $\bar{\boldsymbol{u}}$ to possess AMP $\boldsymbol{\alpha}$ is estimated by

$$P\left(\boldsymbol{\alpha}|\bar{\boldsymbol{u}}\right) = P\left(\bar{\boldsymbol{u}}|\boldsymbol{\alpha}\right)P(\boldsymbol{\alpha})/P(\bar{\boldsymbol{u}}), \tag{17}$$

where $P\left(\bar{\boldsymbol{u}}|\boldsymbol{\alpha}\right)$ is the conditional probability for working out the response vector $\bar{\boldsymbol{u}}$ given AMP $\boldsymbol{\alpha}$, $P(\boldsymbol{\alpha})$ is the prior probability of $\boldsymbol{\alpha}$, and $P(\bar{\boldsymbol{u}})$ is the probability of getting $\bar{\boldsymbol{u}}$.

Step 2 - Prediction of responses to a tree branch: For selecting the $(m+1)^{\text{th}}$ item, an H-horizon branch $\boldsymbol{b}$ is grown from the constructing test $\bar{\boldsymbol{x}}$, where H is the predefined search depth. Let $\boldsymbol{e}$ denote the potential path that comprises $\bar{\boldsymbol{x}}$ and $\boldsymbol{b}$, i.e., $\boldsymbol{e} = \bar{\boldsymbol{x}} + \boldsymbol{b}$. Then a possible response list to $\boldsymbol{e}$ can be written as $\bar{\boldsymbol{w}} = \bar{\boldsymbol{u}} + \boldsymbol{v}$, in which the front part $\bar{\boldsymbol{u}}$ denotes the actual response vector to $\bar{\boldsymbol{x}}$ and the latter part $\boldsymbol{v}$ is a possible response vector to $\boldsymbol{b}$. Based on the search state obtained at *Step 1*, the probability of getting the response vector $\boldsymbol{v}$, which is also the probability of getting the response list $\bar{\boldsymbol{w}}$, can be computed by

$$P\left(\bar{\boldsymbol{w}}\right) = P(\boldsymbol{v}) = \sum\nolimits_{\boldsymbol{\alpha}} P(\boldsymbol{v}|\boldsymbol{\alpha})P(\boldsymbol{\alpha}|\bar{\boldsymbol{u}}), \tag{18}$$

where $P(\boldsymbol{v}|\boldsymbol{\alpha})$ is computed by the IRFs of the CDM in use.

Step 3 - Estimation of the confidence for predicting the examinee's AMP: For each possible response list $\bar{\boldsymbol{w}}$, the posterior probability $P\left(\boldsymbol{\alpha}|\bar{\boldsymbol{w}}\right)$ of AMP $\boldsymbol{\alpha}$ can be computed by Eq. (17) with $\bar{\boldsymbol{w}}$ in place of $\bar{\boldsymbol{u}}$. The AMP with the highest posterior probability is most likely the true state of the examinee. We can thus use the maximal posterior probability among AMPs, i.e., $\max_{\boldsymbol{\alpha}} P\left(\boldsymbol{\alpha}|\bar{\boldsymbol{w}}\right)$, to indicate the confidence of the path $\boldsymbol{e}$ for identifying the examinee's true state in the case of the response list $\bar{\boldsymbol{w}}$. Combining all the possible response lists, the expected value of the maximal posterior probability among AMPs, which implies the diagnostic prospect of the branch $\boldsymbol{b}$, is defined as the heuristic value. That is,

$$f = \sum\nolimits_{\bar{\boldsymbol{w}}} P(\bar{\boldsymbol{w}})\max\nolimits_{\boldsymbol{\alpha}} P\left(\boldsymbol{\alpha}|\bar{\boldsymbol{w}}\right), \tag{19}$$

As such, a larger heuristic value indicates higher confidence to predict the examinee's AMP.

Heuristic-search-based item selection strategies

Pruning progressive heuristic search (PPHS)

Figure 2A displays an illustration of the PPHS strategy, which iteratively expands branches by adding one horizon per iteration. When expanding a branch $\boldsymbol{b}$ at the mth horizon ($m \geq 0$), PPHS starts from exploring all the feasible items in the bank, i.e., items that have not been used and using these items in the test is compliant with the test constraints. Then each feasible item is added to $\boldsymbol{b}$, forming a group of new branches reaching the $(m+1)^{\text{th}}$ horizons. These new branches are evaluated by the heuristic function f defined in Eq. (19). A pruning

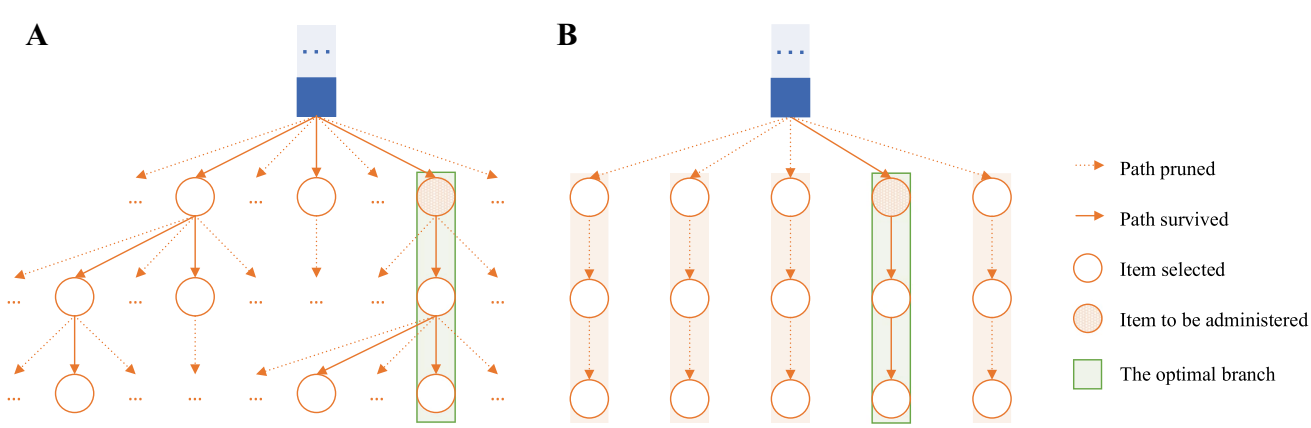

Fig. 2 Illustration of the proposed item selection strategies: **A** PPHS and **B** MCHS

operation is conducted to omit sub-optimal branches, leaving only the best S branches with S as the predefined search width. The above operations are performed for each existing m-horizon branch in the search tree. Then the same procedure is adopted to further grow the branches until the predefined search depth H is reached. The item on the first horizon of the branch that achieves the highest heuristic value in the last iteration is selected as the next item in the test. Notably, PPHS considers all the eligible items but only keeps the few ones that are found the best by the time they are grown. Such a pruning operation thus restricts the exploring direction and lets the heuristic search focus on the promising paths. Meanwhile, the item selection is based on prospect evaluation of the branches reaching the search depth, implying that the item is selected according to the prospect of the search path that starts from it. Algorithm 1 shows the pseudocodes of the PPHS strategy.

Monte Carlo heuristic search (MCHS)

Considering that PPHS can be time-consuming due to its iterative and traversal nature, we further propose the MCHS strategy, which is a heuristic search process based on *Monte Carlo sampling*. Figure 2B presents an illustration of MCHS, which explores a number of full-length branches (i.e., the number of nodes in the branch meets the predefined search depth H) by sampling items across all the horizons. For each branch, MCHS starts from sampling a feasible item for the first horizon. Then the items for the other horizons are sampled in the same way one after another until the search depth H is reached. By doing so, a total of S branches are sampled and evaluated by the heuristic function, where S is the predefined search width. The item on the first horizon of the branch with the highest heuristic information is selected to be the next item in the test. Unlike PPHS, the MCHS strategy does not need to repeatedly evaluate the heuristic

Input: Constructing test $\bar{\boldsymbol{x}}$, response vector $\bar{\boldsymbol{u}}$, search depth H, search width S
Output: Constructing test $\bar{\boldsymbol{x}}$ with one more item

```
let B be an empty matrix
let b_0 be an empty vector (i.e., branch)
B ← B + b_0
h ← 1
while h ≤ H do
    let Z be an empty set
    for each branch b ∈ B do
        e ← b + x̄
        compute the eligible item pool z for e
        for each item i ∈ z do
            b′ ← b + [i]
            evaluate the heuristic function f of b′ based on x̄ and ū
            Z ← Z ∪ {(b′, f_b′)}
        end for
    end for
    b_1, b_2, …, b_S ← the S branches with the highest f in Z
    B ← [b_1, b_2, …, b_S]
    h ← h + 1
end while
x̄ ← x̄ + [b_1(1)]
```

Algorithm 1 Pruning progressive heuristic search (PPHS) strategy

values of branches during the growing period of the search tree. Instead, it only needs to compute the heuristic values after the full-length branches are sampled. The decrease in computational cost makes MCHS much less time-consuming than PPHS, but probably at the cost of algorithmic performance, especially when the search width is small compared to the bank size. Algorithm 2 presents the pseudocodes of MCHS.

Other components of the proposed strategies

Constraint handling scheme

A common constraint regarding item type is considered in the current study (Lin et al., 2019). The item bank Ω includes T types of items. Each item i ($i = 1, 2, ..., I$) appertains to one and only one type. Indicator $\varsigma_i^t \in \{0, 1\}$ represents the subordination of item i to type t, in which $\varsigma_i^t = 1$ indicates that item i belongs to type t and $\varsigma_i^t = 0$ indicates the opposite. For each test $\boldsymbol{x}$ with a predefined test length N, the items' type proportion should meet the required range $[L^t, U^t]$:

$$L^t \leq \frac{1}{N} \sum_{i=1}^{N} \varsigma_{x_i}^t \leq U^t, t = 1, 2, \ldots T. \tag{20}$$

As CD-CAT is assembled in an online manner, the following constraint handling scheme is adopted to assure the legality of the eventual test. For the constructing test $\bar{\boldsymbol{x}}$ with m items, let $\boldsymbol{\mu}$ be the set of item types that have not met the corresponding lower boundaries in the type-proportion constraint and $\boldsymbol{\nu}$ be the set of item types that have reached the corresponding upper boundaries. That is,

$$\boldsymbol{\mu} = \left\{ t : \frac{1}{N} \sum_{i=1}^{m} \varsigma_{\bar{x}_i}^t < L^t \right\}, \tag{21}$$

$$\boldsymbol{\nu} = \left\{ t : \frac{1}{N} \sum_{i=1}^{m} (\varsigma_{\bar{x}_i}^t + 1) > U^t \right\}. \tag{22}$$

If $\boldsymbol{\mu}$ is not empty, the eligible item pool $\boldsymbol{e}$ is restricted to the unused items of types within $\boldsymbol{\mu}$. Otherwise, the unused

Input: Constructing test $\bar{\boldsymbol{x}}$, response vector $\bar{\boldsymbol{u}}$, search depth H, search width S
Output: Constructing test $\bar{\boldsymbol{x}}$ with one more item

```
let Z be an empty set
s ← 1
while s ≤ S do
    b ← []
    e ← x̄
    h ← 1
    while h ≤ H do
        compute the eligible item pool z for e
        randomly choose an item i from z
        b ← b + [i]
        e ← e + [i]
        h ← h + 1
    end while
    evaluate the heuristic function f of b based on x̄ and ū
    Z ← Z ∪ {(b, f_b)}
    s ← s + 1
end while
b_1 ← the optimal branch with the highest heuristic function value in Z
x̄ ← x̄ + [b_1(1)]
```

Algorithm 2 Monte Carlo heuristic search (MCHS) strategy

items that are not of types in $\boldsymbol{v}$ constitute the eligible item pool $\boldsymbol{e}$. By doing so, the ultimate test must satisfy the type-proportion constraint.

First item selection scheme

In practice, there is usually no prior information of the examinee when assigning the first item. However, the proposed item selection strategies, as many others in the literature, rely on the information that is acquired from the examinee's response to the first item to implement. It is thus critical to use a high-quality item as the first item to provide richer information on the examinee and pave the way for the following test. Considering this, we involve the attribute-level discrimination index (ADI) (Henson et al., 2008) to select the first item. ADI is one of the most popular item discrimination indices of CDMs, which measures the diagnostic usefulness of an item between the AMP pairs that differentiate each other by only one attribute. The computation of ADI is also convenient, as it only requires knowing the Q-vector and the parameters of each item.

Specifically, the usefulness for item i to distinguish a master from a non-master or vice versa on attribute k can be described as

$$ADI_{ik} = 2^{-K} \sum\nolimits_{\Delta_k} KL_i(\boldsymbol{\alpha}, \boldsymbol{\alpha'}), \tag{23}$$

where

$$\Delta_k = \left\{ \left(\boldsymbol{\alpha}, \boldsymbol{\alpha'} \right) : \alpha_k \neq \alpha'_k \wedge \alpha_j = \alpha'_j \text{ for } \forall j \neq k \right\}. \tag{24}$$

In this study, the ADI of item i is set as the maximum among the ADI values of all the attributes, i.e., $ADI_i = \max_k ADI_{ik}$, so as to promise the item's strong-point of diagnostic ability. Considering both the quality and variability for test initialization, we adopt the "randomesque" rule (Kingsbury & Zara, 1989) to randomly select the first item from the top 20% of items with the highest ADI in the bank.

Simulation study

To verify the effectiveness of the proposed PPHS and MCHS strategies, a series of simulation studies are performed. We first conduct a parameter investigation to find out the generally appropriate settings for the two proposed strategies. Then two simulation studies that consider different scenarios are performed based on item banks with different designs and quality conditions under different prior distributions of AMPs. The performances of the proposed strategies are compared with the classic item selection strategies in CD-CAT to further illustrate their advantage.

Simulation design

Simulation scenarios

Two scenarios are considered in our simulation studies. The first scenario covers the conventional situation, in which the responses of examinees depend on the IRFs of the CDM in use. The second scenario is designed to further investigate the performances of different strategies when the impact of item familiarity caused by item exposure is taken into account.

Given an item i, we hypothesize that the probability of examinees being familiar with the item is

$$P_{\text{familiar}} = \min\left[\theta\left(\varrho_i / o - 1\right), \beta\right], \tag{25}$$

where ϱ_i is the exposure rate of item i, o is the predefined threshold for overexposure and is set at 0.2 as in previous literature (Zheng & Wang, 2017), $\theta \in \mathbb{R}^+$ and $\beta \in (0, 1)$ are two predefined parameters for controlling the familiarity impact. Then the probability of examinees correctly responding to item i is redefined as

$$P = \begin{cases} 1, & \text{if } \delta < P_{\text{familiar}} \\ P\left(\sigma_i = 1 | \boldsymbol{\alpha}\right), & \text{otherwise} \end{cases}, \tag{26}$$

where $\delta \in [0, 1)$ is a random number generated from the uniform distribution. Once ϱ_i exceeds o, examinees have a P_{familiar} possibility of responding correctly to the item regardless of their knowledge status. P_{familiar} increases as the item continues to be exposed, which simulates the situation that examinees may know or guess the right answers easily to highly exposed items. In the simulations, we set β at 0.8 to impose a moderate upper limit on the probability for examinees to directly make the right response. For θ, we consider two different settings, i.e., $\theta = 0.3$ and $\theta = 0.5$, to simulate a low and a high level of item familiarity impact.

Simulation materials

Two types of materials are needed for our simulation studies: synthetic item banks and virtual examinees whose AMPs are already known. The synthetic banks are generated following the bank designs frequently used in previous empirical and/or simulation studies. For the number of attributes, it has been found that four is the most seen setting in empirical studies (Sessoms & Henson, 2018), while five is the most tested setting in simulations (Nájera et al., 2021). Therefore, we considered two designs of item banks, namely bank A and bank B, which measure four and five attributes with 200

and 300 items, respectively. The test length is set as $N = 32$ for bank A and $N = 40$ for bank B given that eight has been considered a sufficient item to attribute ratio in previous literature (Nájera et al., 2021). For the Q-vector complexity (i.e., percentages of items measuring different number of attributes), the two bank designs use the setting that is recommended by a recent summary over CDM applications (Nájera et al., 2021), as shown in Table 1. Yet the existing literature lack support for setting the proportions of different item types [i.e., multiple-choice (MC), bank-filling (BF), and short essay (SE)]. We then used a uniform setting, with which the proportion of each item type is about 1/3 and the number of required attributes increases along MC, BF, and SE as in real tests. Within each test, the item proportions of MC, BF, and SE are constrained to the ranges of [0.3, 0.4], [0.3, 0.4], and [0.2, 0.3], respectively.

The items in each bank are generated based on the widely used model r-RUM, which, as introduced in "Cognitive diagnostic tests and their computerized adaptive form" section, requires setting a baseline parameter $\widehat{\pi}_i$ and a penalty parameter $\widehat{r}_{ik}$ per attribute for each item i. In literature, it has been suggested that $\widehat{\pi}_i \in U(0.75, 0.95)$ and $\widehat{r}_{ik} \in U(0.2, 0.95)$ is a reasonable and frequently used setting (Henson & Douglas, 2005). To further study performance of the proposed strategies on item banks with different quality, we vary the variability and discrimination of items by manipulating the distributions of item parameters (Kaplan et al., 2015), which are $\widehat{\pi}_i$ and $\widehat{r}_{ik}$ in the current simulation. For item variability, we keep the above $\widehat{\pi}_i$ setting as a high variability level [HV: $\widehat{\pi}_i \in U(0.75, 0.95)$] and narrow it down symmetrically for a low variability level [LV: $\widehat{\pi}_i \in U(0.8, 0.9)$]. For item discrimination, we extract the lower part of the original $\widehat{r}_{ik}$ setting as a high discrimination level [HD: $\widehat{r}_{ik} \in U(0.2, 0.65)$] and use the higher part as a low discrimination level [LD: $\widehat{r}_{ik} \in U(0.5, 0.95)$]. Consequently, the quality of item banks has a two-by-two design ({LD, HD} × {LV, HV}), resulting in a total of four quality conditions, as listed in Table 1.

As for virtual examinees, as suggested in the literature, their true AMPs are randomly drawn from either the uniform distribution (UNI) or the multivariate normal distribution (MVN) (Henson & Douglas, 2005; Finkelman et al., 2009). The UNI distribution simulates the situation that all attributes are uncorrelated, while the MVN distribution assumes the correlation between each attribute pair to be 0.5. The responses of virtual examinees to items in the banks are determined in advance based on the IRFs of r-RUM. Then the diagnostic error rates for different item selection strategies are computed using the same set of virtual examinees. Taken together, a total of 16 conditions (2 bank designs × 4 bank-quality conditions × 2 AMP prior distributions) are established for each of the two scenarios in the simulation studies.

Strategies in comparison and evaluation criteria

In this study, the two proposed item selection strategies are compared with five other strategies, including the random baseline (RND) and four classic item selection strategies in the CD-CAT literature. The four classic strategies fall into two categories: focusing only on diagnostic performance (i.e., SHE and GDI, namely diagnosis-oriented strategies) and considering both diagnostic performance and item exposure control (i.e., RP_PWKL and DBS, namely control-embedded strategies). For the sake of fairness, all strategies adopt the ADI-based first item selection scheme as well as the proposed constraint handling scheme.

Two types of evaluation criteria are employed in the current study. The first type of criteria measures the diagnostic performance, including the attribute- and pattern-level diagnostic error rates, which are obtained by comparing the estimated AMPs and true AMPs across all virtual examinees. The estimated AMPs are computed according to the principle of *Maximum a Posteriori*. Specifically, for each AMP $\widehat{\boldsymbol{\alpha}}$, an exclusive probability is computed by

Table 1 Summary of bank designs and bank-quality conditions

Bank	K	I	Q-vector complexity					Item parameters		
			q_1	q_2	q_3	q_4	q_5	Condition	Baseline ($\widehat{\pi}_i$)	Penalty ($\widehat{r}_{ik}$)
A	4	200	35%	43%	14%	8%	-	LD-LV	$U(0.8, 0.9)$	$U(0.5, 0.95)$
								LD-HV	$U(0.75, 0.95)$	$U(0.5, 0.95)$
								HD-LV	$U(0.8, 0.9)$	$U(0.2, 0.65)$
								HD-HV	$U(0.75, 0.95)$	$U(0.2, 0.65)$
B	5	300	35%	42%	14%	7%	2%	LD-LV	$U(0.8, 0.9)$	$U(0.5, 0.95)$
								LD-HV	$U(0.75, 0.95)$	$U(0.5, 0.95)$
								HD-LV	$U(0.8, 0.9)$	$U(0.2, 0.65)$
								HD-HV	$U(0.75, 0.95)$	$U(0.2, 0.65)$

K: number of attributes; I: size of item bank; q_k: percentage of items measuring k attributes; LD/HD: low/high item discrimination level; LV/HV: low/high item variability level; U: uniform distribution

$$\tau_{\hat{\alpha}} = \sum_{\alpha} \left(\sum_{k} \left| \alpha_k - \hat{\alpha}_k \right| \right) P(\boldsymbol{\alpha}|\boldsymbol{w}). \tag{27}$$

The AMP $\hat{\boldsymbol{\alpha}}$ with the minimal $\tau_{\hat{\alpha}}$ is adopted as the estimated AMP. The second type of evaluation criterion is about the item bank utilization, including the scaled χ^2 (Chang & Ying, 1999) that measures the variance of item exposure rates as

$$\chi^2 = \sum_{i} (o_i - \overline{o})^2 / \overline{o} \text{ with } \overline{o} = N/I, \tag{28}$$

the rate of underused items whose administering frequencies are less than 0.02, and the rate for overused items whose administering frequencies are beyond 0.2 (Zheng & Wang, 2017).

All the strategies in comparison are implemented in Python 3. Unless stated otherwise, the simulation studies are performed on Tianhe-II Supercomputer (Intel Xeon Gold 6150 Processor) through remote deployment. To reduce bias induced by random factors in the item selection strategies, the performance comparison is made based on the results from 16 independent runs, each using a set of 640 examinees randomly drawn from UNI or MVN. As will be seen, the results under the two AMP prior distribution are mostly consistent. For conciseness, we mainly report the results for MVN in the main text. The corresponding results for UNI are presented in the Supplementary Material (SM).

Parameter investigation

In the proposed item selection strategies, the search depth and width determine the scale of the search tree and thus have a significant impact on the results of the heuristic search mechanism. A search depth or width that is too small may reduce the effectiveness of the heuristic search, while an oversize search depth or width may cause excessive search efforts and impair the practicality of CD-CAT as a real-time testing tool. Considering that the scale of the item banks (i.e., bank size and attribute number) is the most influential factor for the search space of the next item, we perform a parameter investigation on banks A and B with the quality condition set as HD-HV and the AMP prior distribution set as MVN. The setting of search depth is first investigated in the range of {1, 2, 3} with the search width fixed at 3 for PPHS and at 200 (or 300) for MCHS for bank A (or B). Then, with the search depth fixed at its best value, the search width of PPHS and MCHS is investigated in the ranges of {1, 2, …, 5} and {100, 200, …, 500}, respectively.

Tables 2 and 3 show the results of PPHS and MCHS under different parameter combinations, respectively. For both strategies on both banks, the diagnostic accuracy significantly improves when the search depth H rises from 1 to 2. Nevertheless, as H climbs up to 3, the diagnostic accuracy no longer significantly increases and even declines in some cases, probably because the estimation bias accumulates and amplifies as the search branch grows in depth, resulting in misleading item selection. Regarding the item bank utilization, increasing H sometimes causes inferior utilization (higher χ^2 and item overused rate), especially for PPHS. As for the search width S of PPHS, the diagnostic accuracy significantly improves as S rises from 1 to 3 but shows little improvement when S rises from 3 to 5. The overall situation of item bank utilization is on the downtrend as the search widens (higher χ^2 and item overused rate). With respect to MCHS, in the case of bank A, $S = 200$ is superior in diagnostic accuracy but inferior in item bank utilization to $S = 100$. Meanwhile, $S = 200$ outperforms $S = 300$ significantly in terms of item bank utilization, while no significant improvement is observed in diagnostic accuracy as S increases. For bank B, $S = 300$ reports significantly better diagnostic accuracy but inferior bank utilization compared to $S = 100$ and 200. Further, $S = 300$ is significantly better than $S = 400$ and 500 in bank utilization while shows no significant difference in diagnostic accuracy. Figure 3 presents the computational time of the two strategies across different parameter settings. As can be seen, on both banks, the time cost increases super-linearly as the search deepens but grows more linearly as the search widens.

Taking both algorithmic performance and computational time into account, we consider $H = 2$ a promising setting for both PPHS and MCHS on both banks. As for the search width, PPHS can use $S = 3$ as a generally good setting, while MCHS is more sensitive to the bank scale, favoring $S = 200$ on bank A but $S = 300$ on bank B. These settings are also employed in the subsequent simulation studies.

Simulation results in scenario 1

Overall comparison

Tables 4 and 5 present the scenario 1 results of the seven item-selection strategies on banks A and B, given that the prior distribution of AMPs is subject to MVN. The corresponding results with AMPs subject to UNI are shown in Tables S1 and S2 in the Supplementary Material (SM).

In general, the rankings of the seven strategies in terms of diagnostic performance and bank utilization align across different bank designs and quality conditions. The baseline RND achieves well-balanced bank utilization, as reflected by χ^2 less than one and almost zero underused/overused rates across all conditions, but its diagnostic error rates are the highest at both attribute- and pattern-levels. As for the two classic diagnosis-oriented strategies (i.e., SHE and GDI), their diagnostic error rates are significantly better than the classic control-embedded strategies (i.e., RP_PWKL

Table 2 Parameter investigation for PPHS

Bank	Indicator (mean ± SD)	Search depth (H) with search width (S) fixed at 3			Search width (S) with search depth (H) fixed at 2				
		$H = 1$	$H = 2$	$H = 3$	$S = 1$	$S = 2$	$S = 3$	$S = 4$	$S = 5$
A	A1	0.020 ± 0.006*	**0.013 ± 0.005**	0.014 ± 0.006	0.018 ± 0.004*	0.016 ± 0.005	0.013 ± 0.005	**0.013 ± 0.003**	0.013 ± 0.004
	A2	0.020 ± 0.004*	**0.013 ± 0.005**	0.015 ± 0.004	0.017 ± 0.003*	0.014 ± 0.003	0.013 ± 0.005	0.012 ± 0.003	**0.011 ± 0.003**
	A3	0.018 ± 0.005*	0.010 ± 0.004	**0.009 ± 0.004**	0.015 ± 0.005*	0.012 ± 0.004	0.010 ± 0.004	**0.008 ± 0.003**	0.010 ± 0.003
	A4	0.013 ± 0.005*	**0.007 ± 0.004**	0.008 ± 0.004	0.009 ± 0.003*	0.009 ± 0.004	0.007 ± 0.004	0.009 ± 0.004	**0.007 ± 0.002**
	Pattern	0.065 ± 0.009*	**0.040 ± 0.006**	0.043 ± 0.009	0.055 ± 0.008*	0.049 ± 0.007*	0.040 ± 0.006	**0.039 ± 0.007**	**0.039 ± 0.007**
	χ^2	**13.875 ± 0.252***	14.541 ± 0.284	15.056 ± 0.301*	13.723 ± 0.403*	**13.615 ± 0.295***	14.541 ± 0.284	14.964 ± 0.375*	15.923 ± 0.246*
	Overused	**0.160 ± 0.012***	0.189 ± 0.008	0.197 ± 0.007*	**0.157 ± 0.009***	0.183 ± 0.009	0.189 ± 0.008	0.200 ± 0.007*	0.198 ± 0.008*
	Underused	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**
B	A1	0.022 ± 0.006*	0.014 ± 0.004	**0.013 ± 0.004**	0.022 ± 0.007*	0.017 ± 0.005	0.014 ± 0.004	**0.013 ± 0.004**	**0.013 ± 0.004**
	A2	0.018 ± 0.005*	**0.011 ± 0.004**	0.012 ± 0.004	0.018 ± 0.004*	0.013 ± 0.004	0.011 ± 0.004	0.010 ± 0.004	**0.010 ± 0.003**
	A3	0.014 ± 0.005*	0.009 ± 0.004	**0.008 ± 0.004**	0.012 ± 0.004*	0.010 ± 0.005	0.009 ± 0.004	**0.008 ± 0.003**	0.009 ± 0.004
	A4	0.014 ± 0.004*	**0.008 ± 0.003**	0.010 ± 0.003	0.013 ± 0.004*	0.011 ± 0.004*	**0.008 ± 0.003**	**0.008 ± 0.003**	**0.008 ± 0.003**
	A5	0.014 ± 0.005*	0.008 ± 0.004	**0.007 ± 0.003**	0.011 ± 0.005*	0.007 ± 0.004	0.008 ± 0.004	**0.007 ± 0.003**	0.007 ± 0.004
	Pattern	0.076 ± 0.010*	0.048 ± 0.008	**0.047 ± 0.009**	0.071 ± 0.012*	0.057 ± 0.007*	0.048 ± 0.008	**0.043 ± 0.009**	0.044 ± 0.008
	χ^2	**20.704 ± 0.458***	23.255 ± 0.340	24.412 ± 0.449*	**20.607 ± 0.340***	21.660 ± 0.384*	23.255 ± 0.340	23.121 ± 0.361	24.244 ± 0.390*
	Overused	0.114 ± 0.003*	0.129 ± 0.005	0.125 ± 0.006*	**0.114 ± 0.002***	0.130 ± 0.005	0.129 ± 0.005	0.133 ± 0.006	0.136 ± 0.007*
	Underused	**0.000 ± 0.000**	0.000 ± 0.001	**0.000 ± 0.001**	**0.000 ± 0.000**	**0.000 ± 0.000**	0.000 ± 0.001	0.001 ± 0.001	**0.000 ± 0.001**

H: search depth; S: search width; Ak: attribute-level diagnostic error rate on attribute k; Pattern: pattern-level diagnostic error rate; χ^2: variance of item exposure rate; Overused/Underused: rates of overused/underused items; '*': significant difference with the chosen parameters ($H = 2$ and $S = 3$) at the 95% confidence level according to the Wilcoxon rank-sum test; **bold** number: best results among the parameter settings in comparison

and DBS) and the two proposed strategies (i.e., PPHS and MCHS). Nevertheless, the diagnosis-oriented strategies suffer from poor bank utilization. Their χ^2 values and underused/overused rates are significantly worse than the other strategies with only one exception: the item overused rate of GDI is significantly better than DBS and even MCHS in some cases but remains inferior to PPHS. The classic control-embedded strategies (i.e., RP_PWKL and DBS) consistently report significantly worse diagnostic accuracy than the proposed PPHS and MCHS. It is only under the HD conditions on bank B that RP_PWKL and DBS can achieve diagnostic error rates matchable to PPHS, but are still inferior to MCHS. However, RP_PWKL achieves excellent bank utilization, as reflected by the smallest χ^2 values among all non-random methods and zero item overused/underused rates. DBS has the second best χ^2 values but exhibits a higher item overused rate than the proposed PPHS and MCHS in most cases, showing problematic item overexposure control. Between the proposed strategies PPHS and MCHS, their diagnostic performance shows no significant difference in most cases. Both PPHS and MCHS maintain their item underused rates close to zero throughout different simulation conditions, which indicates their excellent ability to minimize item waste. MCHS consistently outperforms

Table 3 Parameter investigation for MCHS

Bank	Indicator (mean ± SD)	Search depth (H) with search width (S) fixed at 200 (Bank A) or 300 (Bank B)			Search width (S) with search depth (H) fixed at 2				
		$H = 1$	$H = 2$	$H = 3$	$S = 100$	$S = 200$	$S = 300$	$S = 400$	$S = 500$
A	A1	0.019 ± 0.005*	**0.012 ± 0.002**	0.013 ± 0.005	0.015 ± 0.004*	0.012 ± 0.002	0.014 ± 0.004	0.012 ± 0.004	**0.010 ± 0.004**
	A2	0.020 ± 0.004*	0.014 ± 0.004	**0.011 ± 0.005**	0.016 ± 0.005	0.014 ± 0.004	**0.011 ± 0.004**	0.014 ± 0.004	0.013 ± 0.004
	A3	0.015 ± 0.004*	0.009 ± 0.004	**0.008 ± 0.003**	0.012 ± 0.003*	0.009 ± 0.004	**0.009 ± 0.002**	0.009 ± 0.003	0.010 ± 0.003
	A4	0.012 ± 0.004*	**0.009 ± 0.004**	**0.009 ± 0.004**	0.009 ± 0.003	0.009 ± 0.004	**0.008 ± 0.003**	**0.008 ± 0.003**	0.008 ± 0.004
	Pattern	0.060 ± 0.009*	0.041 ± 0.005	**0.039 ± 0.009**	0.049 ± 0.007*	0.041 ± 0.005	**0.038 ± 0.006**	0.039 ± 0.006	0.039 ± 0.005
	χ^2	11.494 ± 0.247*	10.737 ± 0.296	**10.117 ± 0.228***	**8.873 ± 0.304***	10.737 ± 0.296	12.030 ± 0.247*	12.790 ± 0.327*	13.387 ± 0.375*
	Overused	**0.177 ± 0.009***	0.227 ± 0.010	0.233 ± 0.009	**0.222 ± 0.011**	0.227 ± 0.010	0.227 ± 0.008	0.225 ± 0.006	0.226 ± 0.006
	Underused	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**
B	A1	0.020 ± 0.005*	0.011 ± 0.004	**0.010 ± 0.004**	0.016 ± 0.004*	0.012 ± 0.005	**0.011 ± 0.004**	0.012 ± 0.004	0.013 ± 0.005
	A2	0.018 ± 0.004*	0.010 ± 0.003	**0.008 ± 0.004**	0.012 ± 0.004	0.011 ± 0.003	0.010 ± 0.003	**0.009 ± 0.004**	**0.009 ± 0.004**
	A3	0.015 ± 0.004*	**0.007 ± 0.002**	**0.007 ± 0.002**	0.009 ± 0.003*	0.008 ± 0.003	0.007 ± 0.002	**0.006 ± 0.002**	0.006 ± 0.003
	A4	0.013 ± 0.002*	0.007 ± 0.003	**0.007 ± 0.002**	0.009 ± 0.004	0.009 ± 0.004	0.007 ± 0.003	0.008 ± 0.003	**0.006 ± 0.003**
	A5	0.010 ± 0.003*	**0.006 ± 0.002**	0.006 ± 0.003	0.009 ± 0.003*	0.008 ± 0.004	**0.006 ± 0.002**	0.006 ± 0.003	0.006 ± 0.003
	Pattern	0.072 ± 0.009*	0.039 ± 0.006	**0.037 ± 0.006**	0.052 ± 0.010*	0.046 ± 0.006*	**0.039 ± 0.006**	0.039 ± 0.008	0.039 ± 0.008
	χ^2	17.077 ± 0.310*	15.705 ± 0.331	**14.841 ± 0.373***	**11.384 ± 0.265***	14.061 ± 0.271*	15.705 ± 0.331	16.604 ± 0.255*	17.567 ± 0.368*
	Overused	**0.119 ± 0.006***	0.159 ± 0.007	0.160 ± 0.007	**0.146 ± 0.005***	0.153 ± 0.006*	0.159 ± 0.007	0.162 ± 0.006	0.165 ± 0.007*
	Underused	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**

H: search depth; S: search width; Ak: attribute-level diagnostic error rate on attribute k; Pattern: pattern-level diagnostic error rate; χ^2: variance of item exposure rate; Overused/Underused: rates of overused/underused items; '*': significant difference with the chosen parameters ($H = 2$ and $S = 200/300$) at the 95% confidence level according to the Wilcoxon rank-sum test; **bold** number: best results among the parameter settings in comparison

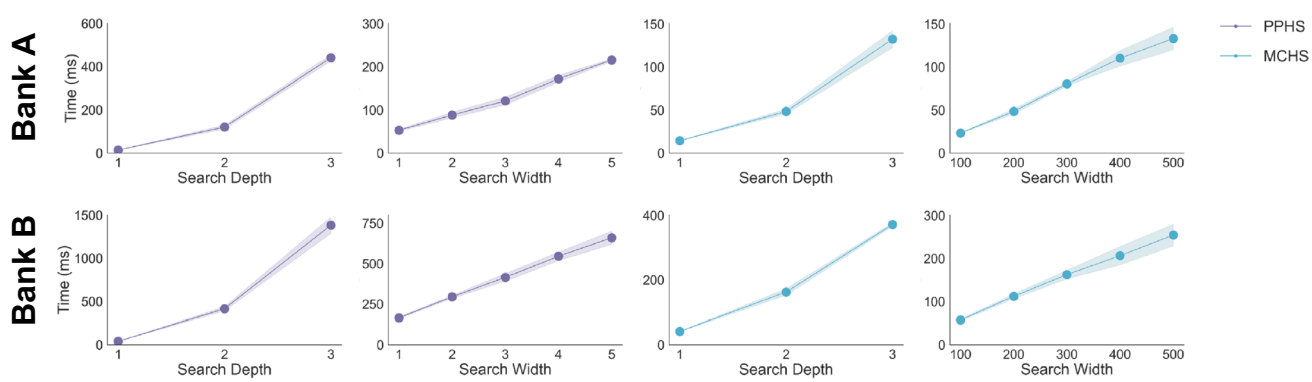

Fig. 3 Computational time of PPHS and MCHS under different parameter settings

Table 4 Performance comparison on Bank A in scenario 1

Bank-Quality Condition	Indicator (mean ± SD)	RND	SHE	GDI	RP_PWKL	DBS	PPHS	MCHS
LD-LV	A1	0.153 ± 0.011*†	**0.060 ± 0.013*†**	0.092 ± 0.012	0.119 ± 0.013*†	0.125 ± 0.012*†	0.086 ± 0.010	0.085 ± 0.017
	A2	0.150 ± 0.010*†	**0.065 ± 0.010*†**	0.095 ± 0.012	0.129 ± 0.009*†	0.124 ± 0.011*†	0.093 ± 0.007	0.089 ± 0.009
	A3	0.136 ± 0.013*†	**0.049 ± 0.007*†**	0.057 ± 0.007*†	0.099 ± 0.013*†	0.110 ± 0.011*†	0.079 ± 0.011	0.074 ± 0.011
	A4	0.133 ± 0.012*†	**0.048 ± 0.008*†**	0.061 ± 0.008*†	0.093 ± 0.010*†	0.119 ± 0.012*†	0.081 ± 0.008	0.076 ± 0.010
	Pattern	0.440 ± 0.012*†	**0.190 ± 0.017*†**	0.262 ± 0.015*	0.343 ± 0.013*†	0.368 ± 0.022*†	0.281 ± 0.014	0.272 ± 0.018
	χ^2	0.620 ± 0.031*†	68.027 ± 0.747*†	102.352 ± 0.564*†	**0.419 ± 0.007*†**	6.556 ± 0.377*†	16.460 ± 0.360†	14.209 ± 0.428*
	Overused	0.031 ± 0.010*†	0.285 ± 0.002*†	0.233 ± 0.005*†	**0.000 ± 0.000*†**	0.306 ± 0.010*†	0.226 ± 0.010†	0.265 ± 0.008*
	Underused	**0.000 ± 0.000**	0.446 ± 0.009*†	0.592 ± 0.007*†	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**
LD-HV	A1	0.152 ± 0.017*†	**0.057 ± 0.009*†**	0.083 ± 0.010	0.113 ± 0.011*†	0.123 ± 0.010*†	0.080 ± 0.008	0.081 ± 0.013
	A2	0.150 ± 0.013*†	**0.054 ± 0.007*†**	0.097 ± 0.013	0.125 ± 0.013*†	0.124 ± 0.013*†	0.091 ± 0.011	0.094 ± 0.009
	A3	0.128 ± 0.011*†	**0.043 ± 0.007*†**	0.055 ± 0.008*†	0.097 ± 0.010*†	0.100 ± 0.008*†	0.070 ± 0.007	0.072 ± 0.011
	A4	0.135 ± 0.013*†	**0.038 ± 0.007*†**	0.050 ± 0.009*†	0.089 ± 0.011*†	0.113 ± 0.009*†	0.067 ± 0.007	0.068 ± 0.008
	Pattern	0.433 ± 0.015*†	**0.166 ± 0.010*†**	0.243 ± 0.012*†	0.334 ± 0.020*†	0.353 ± 0.015*†	0.261 ± 0.013	0.261 ± 0.018
	χ^2	0.614 ± 0.043*†	70.732 ± 0.614*†	102.348 ± 0.573*†	**0.408 ± 0.006*†**	6.371 ± 0.226*†	19.796 ± 0.470†	17.792 ± 0.322*
	Overused	0.035 ± 0.014*†	0.272 ± 0.005*†	0.232 ± 0.004*†	**0.000 ± 0.000*†**	0.300 ± 0.017*†	0.224 ± 0.008†	0.250 ± 0.011*
	Underused	**0.000 ± 0.000**	0.455 ± 0.006*†	0.577 ± 0.004*†	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**
HD-LV	A1	0.059 ± 0.012*†	**0.006 ± 0.003*†**	0.010 ± 0.004*†	0.022 ± 0.004*†	0.018 ± 0.004	0.017 ± 0.005	0.017 ± 0.006
	A2	0.057 ± 0.008*†	**0.006 ± 0.003*†**	0.011 ± 0.004*	0.020 ± 0.005*†	0.017 ± 0.006	0.016 ± 0.005	0.014 ± 0.006
	A3	0.047 ± 0.008*†	**0.004 ± 0.002*†**	0.006 ± 0.003*†	0.017 ± 0.004*†	0.014 ± 0.006	0.012 ± 0.004	0.012 ± 0.003
	A4	0.046 ± 0.006*†	**0.004 ± 0.003*†**	0.007 ± 0.003*†	0.011 ± 0.004	0.015 ± 0.005*†	0.010 ± 0.004	0.011 ± 0.004
	Pattern	0.182 ± 0.016*†	**0.019 ± 0.004*†**	0.030 ± 0.007*†	0.060 ± 0.007*†	0.059 ± 0.006*†	0.051 ± 0.011	0.049 ± 0.010
	χ^2	0.625 ± 0.027*†	49.697 ± 0.612*†	98.399 ± 0.542*†	**0.362 ± 0.007*†**	7.063 ± 0.231*†	14.591 ± 0.378†	10.173 ± 0.239*
	Overused	0.032 ± 0.009*†	0.297 ± 0.004*†	0.241 ± 0.004*†	**0.000 ± 0.000*†**	0.326 ± 0.012*†	0.194 ± 0.006†	0.258 ± 0.008*
	Underused	**0.000 ± 0.000**	0.296 ± 0.009*†	0.552 ± 0.007*†	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**

Table 4 (continued)

Bank-Quality Condition	Indicator (mean ± SD)	RND	SHE	GDI	RP_PWKL	DBS	PPHS	MCHS
HD-HV	A1	0.060 ± 0.007*†	**0.005 ± 0.002*†**	0.009 ± 0.003*†	0.020 ± 0.005*†	0.017 ± 0.005†	0.013 ± 0.005	0.012 ± 0.002
	A2	0.061 ± 0.007*†	**0.004 ± 0.002*†**	0.008 ± 0.004*†	0.021 ± 0.004*†	0.014 ± 0.004	0.013 ± 0.005	0.014 ± 0.004
	A3	0.046 ± 0.007*†	**0.003 ± 0.002*†**	0.005 ± 0.002*†	0.015 ± 0.005*†	0.012 ± 0.004	0.010 ± 0.004	0.009 ± 0.004
	A4	0.046 ± 0.006*†	**0.002 ± 0.002*†**	0.005 ± 0.003†	0.013 ± 0.005*†	0.013 ± 0.004*†	0.007 ± 0.004	0.009 ± 0.004
	Pattern	0.189 ± 0.009*†	**0.013 ± 0.004*†**	0.025 ± 0.005*†	0.059 ± 0.008*†	0.050 ± 0.008*†	0.040 ± 0.006	0.041 ± 0.005
	χ^2	0.630 ± 0.042*†	53.997 ± 0.430*†	98.389 ± 0.728*†	**0.364 ± 0.008*†**	6.857 ± 0.149*†	14.541 ± 0.284†	10.737 ± 0.296*
	Overused	0.042 ± 0.013*†	0.290 ± 0.005*†	0.236 ± 0.005*†	**0.000 ± 0.000*†**	0.323 ± 0.010*†	0.189 ± 0.008†	0.227 ± 0.010*
	Underused	**0.000 ± 0.000**	0.314 ± 0.011*†	0.547 ± 0.006*†	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**

LD/HD: low/high item discrimination; LV/HV: low/high item variability; A*k*: attribute-level diagnostic error rate on attribute *k*; Pattern: pattern-level diagnostic error rate; χ^2: variance of item exposure rate; Overused/Underused: rates of overused/underused items; '*'/'†': significant difference with PPHS/MCHS at the 95% confidence level according to the Wilcoxon rank-sum test; **bold** number: best results among all strategies

PPHS in terms of χ^2, while PPHS holds stable advantage over MCHS in terms of the item overused rate.

Though the overall rankings of the seven strategies are stable, bank design and quality do influence their item selection performance. To investigate the effect of bank scale (i.e., bank size and attribute number), we perform supplementary simulations on two other banks C and D that are different from A and B only by one dimension of the bank scale. Together, the bank scale settings of these four banks form a two-by-two design. The results of the seven strategies on banks C and D are presented in Tables S3–S6. As expected, the overall rankings of the seven strategies remain unchanged on banks C and D. Between each pair of banks that have the same attribute number but different bank sizes, the result comparison shows that increasing the bank size allows the strategies to select from more varied items, which leads to better diagnostic accuracy and overexposure control but results in higher χ^2 and item underused rates. This negative influence on bank utilization is particularly serious on the classic diagnosis-oriented strategies (i.e., SHE and GDI). On the other hand, between each bank pair that have the same size but different numbers of attributes, the comparison shows that increment in the attribute number makes item selection a more difficult problem, resulting in lower diagnostic performance and more serious item overexposure. However, we observe higher χ^2 values and underused rates on the banks that measure fewer attributes, probably because the test length is smaller in this case. Comparisons of the results under the four bank-quality conditions within each bank reveal that low bank quality (low discrimination or/and low variability) negatively impacts diagnostic performance of the non-random strategies, as reflected by the increasing diagnostic error rates in the order of HD-HV, HD-LV, LD-HV, and LD-LV. Notably, under the LD-LV condition, the pattern-level diagnostic error rates of PPHS and MCHS are at approximately 0.3 in most cases and remain below 0.4 in the worst case, demonstrating that the proposed strategies can achieve acceptable diagnostic accuracy even with low-quality banks. Low item discrimination also has a negative impact on bank utilization. Under the LD conditions, all the non-random strategies except for DBS have larger χ^2. The diagnosis-oriented strategies further report higher underused rates, while RP_PWKL, DBS, and the proposed PPHS and MCHS maintain their excellent performance in this regard.

Besides the performance in terms of diagnostic performance and bank utilization, we also compare the average computational time for the seven strategies to select an item. Since the computational time of item selection is mainly influenced by the scale of item banks, the investigation is conducted on banks A and B using the HD-HV condition as an example. To avoid network influence, the

Table 5 Performance comparison on Bank B in scenario 1

Bank-Quality Condition	Indicator (mean ± SD)	RND	SHE	GDI	RP_PWKL	DBS	PPHS	MCHS
LD-LV	A1	0.157 ± 0.013*†	**0.057 ± 0.009*†**	0.082 ± 0.011	0.117 ± 0.012*†	0.117 ± 0.015*†	0.088 ± 0.010	0.086 ± 0.009
	A2	0.138 ± 0.010*†	**0.049 ± 0.007*†**	0.069 ± 0.006*†	0.101 ± 0.009*†	0.114 ± 0.012*†	0.086 ± 0.010	0.081 ± 0.008
	A3	0.141 ± 0.015*†	**0.044 ± 0.007*†**	0.065 ± 0.010*†	0.099 ± 0.010*†	0.113 ± 0.010*†	0.072 ± 0.010	0.075 ± 0.010
	A4	0.147 ± 0.018*†	**0.041 ± 0.006*†**	0.055 ± 0.007*†	0.098 ± 0.012*†	0.116 ± 0.011*†	0.079 ± 0.009	0.077 ± 0.008
	A5	0.148 ± 0.010*†	**0.045 ± 0.007*†**	0.063 ± 0.012*†	0.098 ± 0.011*†	0.111 ± 0.012*†	0.079 ± 0.010	0.079 ± 0.010
	Pattern	0.519 ± 0.015*†	**0.197 ± 0.009*†**	0.276 ± 0.019*†	0.371 ± 0.021*†	0.415 ± 0.017*†	0.324 ± 0.016	0.318 ± 0.013
	χ^2	**0.952 ± 0.035*†**	96.908 ± 0.648*†	153.700 ± 0.737*†	2.032 ± 0.029*†	10.334 ± 0.443*†	24.805 ± 0.558†	19.473 ± 0.561*
	Overused	**0.000 ± 0.000*†**	0.251 ± 0.004*†	0.199 ± 0.003*†	**0.000 ± 0.000*†**	0.196 ± 0.013*†	0.170 ± 0.009†	0.215 ± 0.008*
	Underused	**0.000 ± 0.000**	0.489 ± 0.006*†	0.612 ± 0.005*†	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.001**	**0.000 ± 0.001**
LD-HV	A1	0.161 ± 0.017*†	**0.053 ± 0.007*†**	0.078 ± 0.012*†	0.119 ± 0.013*†	0.113 ± 0.009*†	0.087 ± 0.011	0.090 ± 0.010
	A2	0.140 ± 0.012*†	**0.043 ± 0.007*†**	0.060 ± 0.007*†	0.097 ± 0.009*†	0.111 ± 0.014*†	0.081 ± 0.010	0.077 ± 0.006
	A3	0.132 ± 0.013*†	**0.037 ± 0.005*†**	0.058 ± 0.008*	0.090 ± 0.011*†	0.105 ± 0.010*†	0.067 ± 0.010	0.064 ± 0.009
	A4	0.148 ± 0.018*†	**0.035 ± 0.007*†**	0.050 ± 0.007*†	0.098 ± 0.014*†	0.105 ± 0.012*†	0.069 ± 0.009	0.068 ± 0.009
	A5	0.138 ± 0.014*†	**0.035 ± 0.006*†**	0.055 ± 0.011*†	0.094 ± 0.009*†	0.109 ± 0.010*†	0.068 ± 0.007	0.064 ± 0.010
	Pattern	0.508 ± 0.015*†	**0.173 ± 0.009*†**	0.254 ± 0.019*†	0.368 ± 0.018*†	0.397 ± 0.017*†	0.298 ± 0.013	0.293 ± 0.012
	χ^2	**0.951 ± 0.028*†**	105.491 ± 0.603*†	152.078 ± 0.677*†	2.127 ± 0.020*†	10.137 ± 0.310*†	26.823 ± 0.568†	22.195 ± 0.368*
	Overused	**0.000 ± 0.000*†**	0.235 ± 0.004*†	0.194 ± 0.003*	**0.000 ± 0.000*†**	0.187 ± 0.014*	0.176 ± 0.004†	0.193 ± 0.004*
	Underused	**0.000 ± 0.000**	0.493 ± 0.005*†	0.609 ± 0.006*†	**0.000 ± 0.000**	**0.000 ± 0.000**	0.001 ± 0.003	**0.000 ± 0.001**
HD-LV	A1	0.067 ± 0.008*†	**0.003 ± 0.002*†**	0.004 ± 0.002*†	0.014 ± 0.004	0.011 ± 0.004	0.014 ± 0.004	0.014 ± 0.004
	A2	0.053 ± 0.005*†	**0.003 ± 0.002*†**	0.004 ± 0.002*†	0.012 ± 0.004	0.011 ± 0.004	0.012 ± 0.004	0.010 ± 0.005
	A3	0.049 ± 0.008*†	**0.002 ± 0.002*†**	0.003 ± 0.002*†	0.010 ± 0.004	0.010 ± 0.004	0.011 ± 0.004	0.009 ± 0.003
	A4	0.049 ± 0.009*†	**0.002 ± 0.002*†**	0.003 ± 0.001*†	0.010 ± 0.003	0.011 ± 0.004	0.010 ± 0.003	0.009 ± 0.003
	A5	0.050 ± 0.006*†	**0.002 ± 0.002*†**	0.003 ± 0.002*†	0.011 ± 0.004	0.010 ± 0.004	0.011 ± 0.003†	0.009 ± 0.003*
	Pattern	0.227 ± 0.012*†	**0.012 ± 0.004*†**	0.017 ± 0.005*†	0.049 ± 0.007*	0.048 ± 0.009*	0.055 ± 0.007†	0.048 ± 0.008*
	χ^2	**0.933 ± 0.054*†**	77.102 ± 0.601*†	150.261 ± 1.132*†	1.368 ± 0.019*†	11.210 ± 0.206*†	22.619 ± 0.527†	14.927 ± 0.338*
	Overused	**0.000 ± 0.000*†**	0.246 ± 0.004*†	0.193 ± 0.004*†	**0.000 ± 0.000*†**	0.222 ± 0.012*†	0.154 ± 0.007†	0.183 ± 0.006*
	Underused	**0.000 ± 0.000**	0.373 ± 0.006*†	0.594 ± 0.009*†	**0.000 ± 0.000**	0.001 ± 0.002	**0.000 ± 0.001**	**0.000 ± 0.001**
HD-HV	A1	0.063 ± 0.008*†	**0.003 ± 0.002*†**	0.005 ± 0.003*†	0.016 ± 0.003†	0.012 ± 0.005	0.014 ± 0.004†	0.011 ± 0.004*
	A2	0.046 ± 0.008*†	**0.001 ± 0.002*†**	0.004 ± 0.002*†	0.010 ± 0.003	0.010 ± 0.004	0.011 ± 0.004	0.010 ± 0.003
	A3	0.048 ± 0.008*†	**0.001 ± 0.001*†**	0.002 ± 0.001*†	0.009 ± 0.003†	0.010 ± 0.004	0.009 ± 0.004	0.007 ± 0.002
	A4	0.054 ± 0.009*†	**0.001 ± 0.002*†**	0.002 ± 0.002*†	0.010 ± 0.004†	0.012 ± 0.003*†	0.008 ± 0.003	0.007 ± 0.003
	A5	0.050 ± 0.011*†	**0.001 ± 0.002*†**	0.003 ± 0.002*†	0.010 ± 0.004†	0.010 ± 0.003†	0.008 ± 0.004	0.006 ± 0.002
	Pattern	0.226 ± 0.017*†	**0.008 ± 0.004*†**	0.015 ± 0.003*†	0.048 ± 0.008†	0.048 ± 0.008†	0.048 ± 0.008†	0.039 ± 0.006*
	χ^2	**0.947 ± 0.039*†**	83.505 ± 0.499*†	148.914 ± 0.751*†	1.473 ± 0.016*†	11.059 ± 0.228*†	23.255 ± 0.340†	15.705 ± 0.331*
	Overused	**0.000 ± 0.001*†**	0.237 ± 0.006*†	0.196 ± 0.002*†	**0.000 ± 0.000*†**	0.218 ± 0.008*†	0.129 ± 0.005†	0.159 ± 0.007*
	Underused	**0.000 ± 0.000**	0.380 ± 0.007*†	0.584 ± 0.008*†	**0.000 ± 0.000**	0.002 ± 0.002†	**0.000 ± 0.001**	**0.000 ± 0.000**

LD/HD: low/high item discrimination; LV/HV: low/high item variability; A*k*: attribute-level diagnostic error rate on attribute *k*; Pattern: pattern-level diagnostic error rate; χ^2: variance of item exposure rate; Overused/Underused: rates of overused/underused items; '*'/'†': significant difference with PPHS/MCHS at the 95% confidence level according to the Wilcoxon rank-sum test; **bold** number: best results among all strategies

environment of the investigation is moved to a local computer (Mac Mini with Apple M2 Pro processor). The results are given in Fig. 4 and Fig. S1 in the SM. As expected, the proposed strategies, PPHS and MCHS, need longer computational time than the other strategies due to the tree-based search mechanism. PPHS is even more time-consuming than MCHS, probably because it needs to traverse all feasible items before pruning branches. Nevertheless, the average computational time for the seven strategies is all below half a second, which should be well accepted for real-time testing.

In-depth comparison

In this part of the paper, we aim to further inspect the difference in the performance of the seven item selection strategies in comparison. Since the diagnostic accuracy of RND is far worse than the others across all conditions, it is excluded from the analyses in this part.

To gain a more detailed understanding of item bank utilization, we draw a color grid diagram for each item selection strategy across different simulation conditions, using one of the corresponding 16 runs as an illustrative example. Each grid represents the utilization frequency of an item, with darker colors indicating higher frequencies. The grids are organized in ascending order of the number of attributes measured by the corresponding items, from the top left to the bottom right. Figure 5 displays the diagrams with AMPs subject to MVN, and the corresponding diagrams for UNI are given in Fig. S2. As can be seen, the diagrams of each strategy are similar across different bank designs and quality conditions. Among the six strategies, GDI displays the most polarized utilization, indicating a significant bias towards specific items. Furthermore, GDI demonstrates a preference for items located in the front part of the diagram, suggesting a tendency to select items that measure fewer attributes. SHE exhibits the second most polarized utilization but does not show a clear item preference regarding the number of attributes being measured. On the other hand, the diagrams of RP_PWKL show the most balanced bank utilization among the six strategies, suggesting that RP_PWKL effectively controls item overexposure and underutilization by embedding item usage into the item selection criterion. DBS also demonstrates relatively balanced bank utilization with gentler color distribution. However, it is notable that DBS exhibits a bias towards selecting items from the initial rows in the diagram, indicating a preference for items that measure a single attribute. When compared with the four classic strategies, the diagrams of the two proposed strategies, PPHS and MCHS, exhibit moderately even color distributions with only a few highly exposed items. In summary, although PPHS and MCHS do not achieve as even bank utilization as RP_PWKL and DBS, they offer a novel way for controlling item usage without introducing bias to specific items.

To examine whether the item selection strategies in comparison show bias in diagnosing different AMPs, we investigate their diagnostic error rates across AMPs that master different numbers of attributes. Figure 6 and Fig. S3 tabulate the results with AMPs subject to MVN and UNI, respectively. Among the six strategies, SHE achieves the best diagnostic error rates on most AMPs, usually followed by GDI. The two proposed strategies, PPHS and MCHS, can achieve an accuracy comparable with GDI or even SHE when diagnosing AMPs that master all the attributes. In the MVN case, the diagnostic performance of PPHS and MCHS is also close to SHE and GDI on the empty AMP. On the other hand, PPHS and MCHS demonstrate consistent diagnostic advantage over RP_PWKL and DBS on the majority of AMPs. It is only when diagnosing AMPs that has one or zero attribute can RP_PWKL sometimes achieve a certain advantage. In summary, the above analysis further confirms the overall advantage of PPHS and MCHS over the classic item selection strategies with item exposure control in diagnostic performance. It also reveals the excellent ability of PPHS and MCHS for diagnosing some specific AMPs.

We further examine the efficiency of the six item selection strategies for diagnosing examinees with the following two indicators: (1) the successful rate to reach certain diagnostic confidence level (i.e., the maximum posterior probability of AMPs is beyond a predefined lower bound $P = 0.7, 0.8$, and 0.9), namely the maximum posterior reaching rate (MPRR); (2) the average number of items needed to reach the specified confidence level, namely the maximum posterior reaching length (MPRL). Note that the MPRL is computed based on the tests for the examinees that pass the diagnostic confidence bound. Tables 6 and 7 tabulate the MVN results of MPRR and MPRL across the three confidence levels on banks A and B, respectively (for UNI results refer to Tables S7 and S8). As expected, SHE generally achieves the best efficiency (highest MPRR and smallest MPRL) among the six strategies. GDI is the second-best method in terms of MPRR, but cannot hold consistent advantage over PPHS and MCHS in terms of MPRL. As for RP_PWKL and DBS, PPHS and MCHS exhibit consistent advantage in both terms of MPPR and MPRL, which further validates the advantage of the proposed strategies in diagnostic performance from the viewpoint of testing efficiency.

Further exploration

In this part, we aim to explore whether other item selection criteria can be adapted to the proposed heuristic search framework for further enhancing diagnostic accuracy. Considering that SHE achieves the best diagnostic performance in the above simulations, we develop variants of PPHS and MCHS by replacing the maximum posterior probability in the heuristic function (Eq. 19) with SHE (Eq. 6). The resulting strategies are namely pruning progressive Shannon entropy (PPSHE) and Monte Carlo Shannon entropy (MCSHE), respectively.

Tables 8 and 9 display the results of PPSHE and MCSHE in comparison with SHE, PPHS, and MCHS given AMPs subject to MVN (results for UNI: Tables S9 and S10). The two SHE variants inherit the advantage of SHE in diagnostic efficiency to different degree. More specifically, the attribute- and pattern-level diagnostic error rates of PPSHE

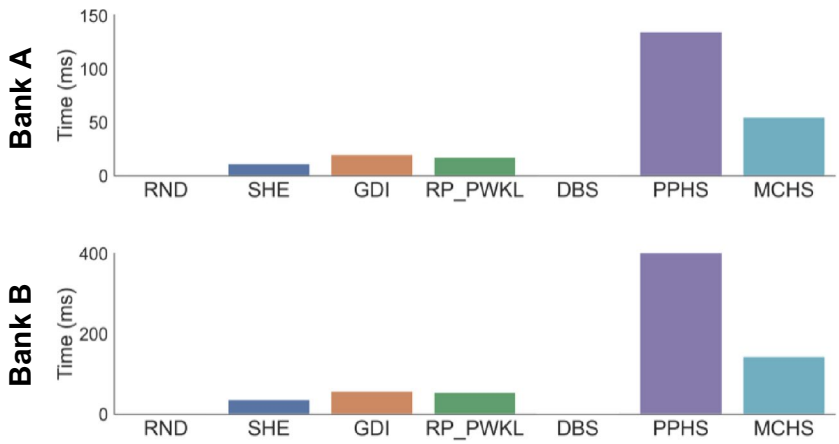

Fig. 4 Comparison on the computational time of item selection strategies

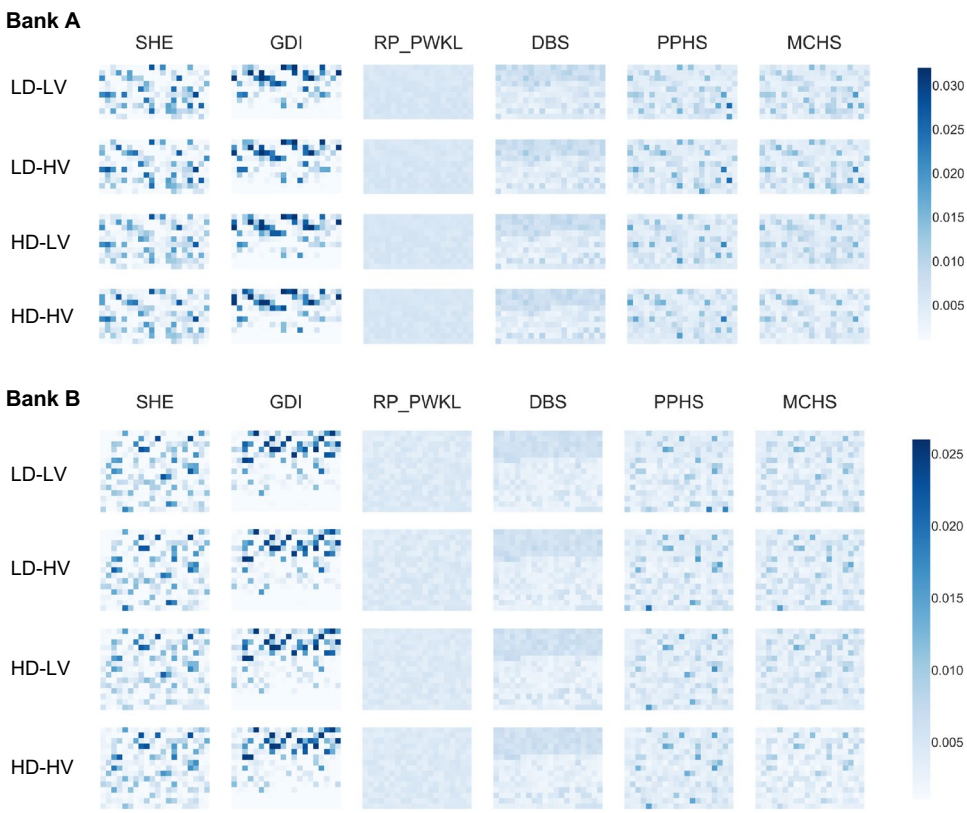

Fig. 5 Comparison on the detailed bank utilization status of item selection strategies

is comparable with SHE on both banks. MCSHE exhibits higher diagnostic error rates than PPSHE and SHE, but still significantly outperforms PPHS and MCHS. However, PPSHE and MCSHE perform worse than PPHS and MCHS in terms of bank utilization. The bank utilization performance of PPSHE is similar to SHE, showing little improvement in the χ^2 value and the item overused/underused rate. MCSHE, on the other hand, outperforms SHE and PPSHE with significantly smaller χ^2 values and item underused rates. However, MCSHE exhibits significant larger item overused rates than SHE and PPSHE, suggesting a limited ability to control item overexposure. Overall, the above results imply that the proposed heuristic search framework is generally effective for enhancing bank utilization. However, the heuristic function should be designed carefully for

achieving the desired balance between diagnostic performance and bank utilization.

Simulation results in scenario 2

The detailed scenario 2 results of the seven item selection strategies across the 16 simulation conditions are tabulated in Tables S11–S18 in the SM. To facilitate summary and comparison, Fig. 7 plots the pattern-level diagnostic error rate (representative indicator for the diagnostic performance) against the χ^2 value (representative indicator for the item bank utilization) for all strategies across both scenarios, given that the prior distribution of AMPs is subject to MVN. To provide additional details and clarity regarding the scenario factor, a figure that depicts the results per scenario is also given in the SM (Fig. S4). The corresponding figures for the UNI case are displayed in Figs. S5 and S6.

As can be seen, the rankings regarding bank utilization for the seven strategies are highly consistent across scenarios, probably because item familiarity is designed to directly affect the IRFs and thus mainly perturbs the diagnostic performance. Regarding diagnostic performance, RND, RP_PWKL, DBS, and the two proposed strategies (PPHS and MCHS) exhibit relatively stable diagnostic accuracy as the level of familiarity impact increases from none (scenario 1) to high (scenario 2, $\theta = 0.5$). In contrast, SHE and GDI are more negatively affected by this impact, resulting in obviously increasing diagnostic error rates. The superiority of SHE and GDI in diagnostic performance thus becomes less evident as the familiarity impact increases. When θ is at 0.3, PPHS and MCHS reverse their disadvantage against GDI in diagnostic accuracy while maintaining their advantage over SHE and GDI in bank utilization across all the 16 simulation conditions. As θ rises to 0.5, the diagnostic accuracy of PPHS and MCHS even surpass SHE on the LD conditions on both banks. Overall, the two proposed strategies gradually become the top two strategies in terms of diagnostic performance as the impact of item familiarity increases. On the LD conditions in scenario 2, they are the only two strategies that can always keep the pattern-level diagnostic error rate below 0.4, while the other strategies surpass 0.4 or even 0.5 in some cases. Meanwhile, their capability in terms of balancing bank utilization stays advantageous over the classic diagnostic-oriented methods across different levels of item familiarity impact.

Discussion and conclusion

The current study proposes to transform the item selection problem in CD-CAT into an optimal path-searching problem on a graph that represents the item bank. The optimization objective is defined as a function that predicts how well the test represented by the path can diagnose the current examinee based on the existing response data. Since a path can involve any numbers of items ahead, this idea enables us to estimate the combined diagnostic utility of two or more items. By doing so, we increase the diversity in item selection when considering diagnostic performance simultaneously. It is thus expected that item bank utilization can be improved at a smaller cost of diagnostic performance. Following the path-searching idea, two item selection strategies,

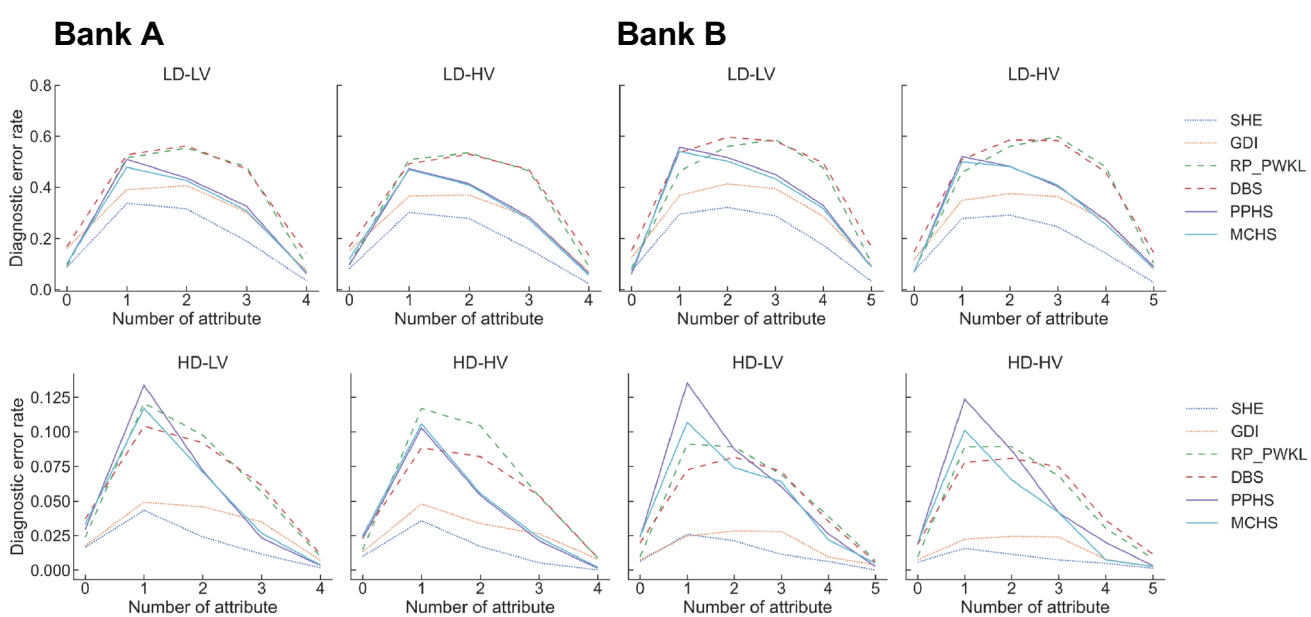

Fig. 6 Comparison on the error rates of item selection strategies for diagnosing AMPs that master different numbers of attributes

Table 6 Efficiency analysis on Bank A

Bank-Quality Condition	*P*	Indicator (mean+SD)	SHE	GDI	RP_PWKL	DBS	PPHS	MCHS
LD-LV	0.7	MPRR	**0.902 ± 0.013*†**	0.800 ± 0.012*†	0.632 ± 0.014*†	0.544 ± 0.015*†	0.694 ± 0.017†	0.728 ± 0.015*
	0.7	MPRL	**13.168 ± 0.244*†**	16.393 ± 0.435*†	18.064 ± 0.371*†	15.453 ± 0.328*†	14.124 ± 0.316†	14.534 ± 0.341*
	0.8	MPRR	**0.840 ± 0.017*†**	0.708 ± 0.015*†	0.539 ± 0.014*†	0.467 ± 0.013*†	0.577 ± 0.020†	0.606 ± 0.014*
	0.8	MPRL	**14.211 ± 0.323*†**	17.811 ± 0.395*†	19.403 ± 0.429*†	15.897 ± 0.419*†	14.790 ± 0.300†	15.103 ± 0.392*
	0.9	MPRR	**0.765 ± 0.017*†**	0.604 ± 0.016*†	0.446 ± 0.013*†	0.393 ± 0.018*†	0.465 ± 0.018†	0.481 ± 0.012*
	0.9	MPRL	15.551 ± 0.272	19.167 ± 0.320*†	20.863 ± 0.431*†	16.426 ± 0.553*†	15.344 ± 0.326	**15.335 ± 0.432**
LD-HV	0.7	MPRR	**0.921 ± 0.012*†**	0.827 ± 0.018*†	0.638 ± 0.020*†	0.568 ± 0.012*†	0.753 ± 0.015	0.755 ± 0.012
	0.7	MPRL	**12.406 ± 0.373*†**	16.037 ± 0.395*†	17.886 ± 0.195*†	15.273 ± 0.423*†	13.705 ± 0.226†	14.133 ± 0.365*
	0.8	MPRR	**0.871 ± 0.011*†**	0.741 ± 0.016*†	0.552 ± 0.021*†	0.492 ± 0.018*†	0.635 ± 0.020	0.644 ± 0.014
	0.8	MPRL	**13.406 ± 0.372*†**	17.483 ± 0.391*†	19.199 ± 0.281*†	15.858 ± 0.386*†	14.550 ± 0.318†	14.862 ± 0.262*
	0.9	MPRR	**0.804 ± 0.013*†**	0.649 ± 0.016*†	0.464 ± 0.018*†	0.415 ± 0.016*†	0.514 ± 0.016†	0.527 ± 0.016*
	0.9	MPRL	**14.601 ± 0.344*†**	18.850 ± 0.433*†	20.652 ± 0.375*†	16.198 ± 0.421*†	15.054 ± 0.417	15.213 ± 0.320
HD-LV	0.7	MPRR	**1.000 ± 0.000**	**1.000 ± 0.001**	0.996 ± 0.003*†	0.995 ± 0.003*†	**1.000 ± 0.001**	**1.000 ± 0.001**
	0.7	MPRL	**5.991 ± 0.123*†**	7.003 ± 0.111†	10.678 ± 0.169*†	9.383 ± 0.174*†	6.950 ± 0.117†	7.679 ± 0.161*
	0.8	MPRR	**1.000 ± 0.001†**	0.999 ± 0.001	0.992 ± 0.003*†	0.990 ± 0.004*†	0.999 ± 0.001	0.998 ± 0.002
	0.8	MPRL	**6.757 ± 0.133*†**	7.901 ± 0.127*†	11.948 ± 0.190*†	10.529 ± 0.189*†	8.261 ± 0.108†	8.725 ± 0.156*
	0.9	MPRR	**0.999 ± 0.002*†**	0.998 ± 0.002*†	0.983 ± 0.004*†	0.980 ± 0.006*†	0.995 ± 0.003	0.995 ± 0.002
	0.9	MPRL	**7.679 ± 0.122*†**	9.222 ± 0.138*†	13.337 ± 0.203*†	11.919 ± 0.193*†	9.726 ± 0.129†	9.934 ± 0.164*
HD-HV	0.7	MPRR	**1.000 ± 0.001**	**1.000 ± 0.000**	0.997 ± 0.003*†	0.995 ± 0.002*†	**1.000 ± 0.000**	**1.000 ± 0.001**
	0.7	MPRL	**5.655 ± 0.139*†**	6.536 ± 0.105*†	10.742 ± 0.230*†	9.206 ± 0.228*†	6.332 ± 0.083†	7.225 ± 0.096*
	0.8	MPRR	**1.000 ± 0.001†**	**1.000 ± 0.001†**	0.992 ± 0.004*†	0.990 ± 0.003*†	0.999 ± 0.001	0.999 ± 0.001
	0.8	MPRL	**6.415 ± 0.131*†**	7.431 ± 0.104†	12.047 ± 0.211*†	10.296 ± 0.217*†	7.502 ± 0.118†	8.220 ± 0.097*
	0.9	MPRR	**0.999 ± 0.001*†**	**0.999 ± 0.002*†**	0.981 ± 0.005*†	0.981 ± 0.005*†	0.997 ± 0.002	0.996 ± 0.002
	0.9	MPRL	**7.111 ± 0.156*†**	8.749 ± 0.105*†	13.369 ± 0.214*†	11.609 ± 0.234*†	8.890 ± 0.149†	9.360 ± 0.087*

LD/HD: low/high item discrimination; LV/HV: low/high item variability; *P*: diagnostic confidence level; MPRR: maximum posterior reaching rate; MPRL: maximum posterior reaching length; '*'/'†': significant difference with PPHS/MCHS at the 95% confidence level according to the Wilcoxon rank-sum test; **bold** number: best results among all strategies

PPHS and MCHS, are developed to approximate the optimal path by implementing a tree-based search mechanism through progressive pruning and Monte Carlo sampling. Simulation results show that the proposed strategies are effective and efficient, offering an innovative way to conduct item selection so that satisfying diagnostic performance and balanced bank utilization can both be achieved.

Merits and costs of the proposed strategies

Given a limited item bank, optimizing diagnostic performance and balancing bank utilization could be two conflicting objectives for test assembly (Lin et al., 2019). Early research on the item selection strategies of CD-CAT focus more on optimizing the diagnostic performance, probably because accuracy is often the first priority at the initial stage of instrument development and CDMs are mostly conceived for low-stakes formative assessments (Cheng, 2009; Henson, 2009). However, as supported by our simulation results, the item selection strategies that are diagnosis-oriented generally suffer from the problems of item overexposure and underutilization. Item overexposure can pose security concerns that lead to threats of test validity (McGlohen & Chang, 2008; Huebner, 2012; Yasuda et al., 2022), while item underutilization can cause waste of bank capacity and result in inefficient investment (van der Linden & Veldkamp, 2004; Wang et al., 2011b). Therefore, the development of item selection strategies with item exposure control are receiving increased attention (Sun et al., 2021; Wang et al., 2011b; Zheng & Wang, 2017). Nevertheless, our simulation results reveal that the existing control-embedded strategies often make a large sacrifice in diagnostic performance for balancing bank utilization. Moreover, recent studies have started to explore the applications of CDTs in high-stakes contexts (Hemati & Baghaei, 2020; Ravand, 2016), which pose a higher demand for both of test accuracy and security. It is thus crucial to develop item selection strategies that can make a good trade-off between diagnostic performance and bank utilization.

Table 7 Efficiency analysis on Bank B

Bank-Quality Condition	*P*	Indicator (mean+SD)	SHE	GDI	RP_PWKL	DBS	PPHS	MCHS
LD-LV	0.7	MPRR	**0.888 ± 0.011***†	0.776 ± 0.018*†	0.575 ± 0.015†	0.474 ± 0.019*†	0.579 ± 0.014†	0.617 ± 0.016*
	0.7	MPRL	**18.415 ± 0.187***†	21.798 ± 0.348*†	24.698 ± 0.446*†	21.805 ± 0.487*†	21.250 ± 0.422	21.108 ± 0.398
	0.8	MPRR	**0.826 ± 0.011***†	0.683 ± 0.019*†	0.490 ± 0.016*	0.402 ± 0.017*†	0.457 ± 0.016†	0.494 ± 0.013*
	0.8	MPRL	**19.789 ± 0.311***†	23.296 ± 0.373*†	26.183 ± 0.373*†	22.563 ± 0.458*†	22.074 ± 0.462	21.753 ± 0.362
	0.9	MPRR	**0.743 ± 0.011***†	0.580 ± 0.016*†	0.399 ± 0.014*†	0.330 ± 0.014†	0.339 ± 0.014†	0.378 ± 0.015*
	0.9	MPRL	**21.182 ± 0.345***†	24.686 ± 0.334*†	27.501 ± 0.410*†	23.309 ± 0.523*†	22.635 ± 0.574†	22.112 ± 0.400*
LD-HV	0.7	MPRR	**0.921 ± 0.011***†	0.808 ± 0.015*†	0.589 ± 0.012*†	0.505 ± 0.015*†	0.641 ± 0.017†	0.659 ± 0.014*
	0.7	MPRL	**17.384 ± 0.263***†	21.137 ± 0.320*†	24.119 ± 0.424*†	21.841 ± 0.553*†	20.340 ± 0.420†	20.798 ± 0.355*
	0.8	MPRR	**0.872 ± 0.011***†	0.723 ± 0.019*†	0.513 ± 0.014†	0.427 ± 0.012*†	0.516 ± 0.015†	0.544 ± 0.017*
	0.8	MPRL	**19.121 ± 0.268***†	22.627 ± 0.448*†	25.611 ± 0.412*†	22.488 ± 0.603*†	21.339 ± 0.418†	21.841 ± 0.313*
	0.9	MPRR	**0.800 ± 0.012***†	0.624 ± 0.016*†	0.421 ± 0.013*	0.351 ± 0.015*†	0.393 ± 0.018†	0.416 ± 0.018*
	0.9	MPRL	**20.473 ± 0.331***†	24.071 ± 0.387*†	27.097 ± 0.394*†	23.125 ± 0.670*†	22.094 ± 0.731	22.266 ± 0.550
HD-LV	0.7	MPRR	**1.000 ± 0.000**	**1.000 ± 0.000**	0.996 ± 0.003*†	0.995 ± 0.003*†	**1.000 ± 0.000**	**1.000 ± 0.001**
	0.7	MPRL	**8.193 ± 0.094***†	9.092 ± 0.140*†	14.608 ± 0.206*†	13.339 ± 0.253*†	10.336 ± 0.143	10.439 ± 0.151
	0.8	MPRR	**1.000 ± 0.000**†	**1.000 ± 0.000**†	0.992 ± 0.003*†	0.991 ± 0.003*†	**1.000 ± 0.001**†	0.999 ± 0.002*
	0.8	MPRL	**9.226 ± 0.110***†	10.424 ± 0.159*†	16.109 ± 0.134*†	14.693 ± 0.293*†	12.110 ± 0.184†	11.772 ± 0.159*
	0.9	MPRR	**1.000 ± 0.001***†	0.999 ± 0.001*†	0.985 ± 0.004*†	0.982 ± 0.007*†	0.996 ± 0.002	0.996 ± 0.003
	0.9	MPRL	**10.390 ± 0.125***†	12.062 ± 0.187*†	17.690 ± 0.144*†	16.259 ± 0.322*†	14.361 ± 0.238†	13.393 ± 0.177*
HD-HV	0.7	MPRR	**1.000 ± 0.000**	**1.000 ± 0.000**	0.997 ± 0.001*†	0.996 ± 0.001*†	**1.000 ± 0.000**	**1.000 ± 0.000**
	0.7	MPRL	**7.394 ± 0.116***†	8.379 ± 0.164*†	14.513 ± 0.201*†	13.037 ± 0.322*†	9.525 ± 0.094†	9.967 ± 0.176*
	0.8	MPRR	**1.000 ± 0.000**	**1.000 ± 0.000**	0.994 ± 0.002*†	0.992 ± 0.003*†	0.999 ± 0.002	0.999 ± 0.001
	0.8	MPRL	**8.294 ± 0.128***†	9.572 ± 0.184*†	16.046 ± 0.195*†	14.390 ± 0.307*†	11.078 ± 0.105†	11.246 ± 0.195*
	0.9	MPRR	**1.000 ± 0.000***†	**1.000 ± 0.001***†	0.988 ± 0.004*†	0.984 ± 0.004*†	0.997 ± 0.003	0.998 ± 0.002
	0.9	MPRL	**9.467 ± 0.158***†	11.201 ± 0.185*†	17.652 ± 0.232*†	15.982 ± 0.290*†	13.126 ± 0.104†	12.777 ± 0.191*

LD/HD: low/high item discrimination; LV/HV: low/high item variability; *P*: diagnostic confidence level; MPRR: maximum posterior reaching rate; MPRL: maximum posterior reaching length; '*'/'†': significant difference with PPHS/MCHS at the 95% confidence level according to the Wilcoxon rank-sum test; **bold** number: best results among all strategies

The simulation results show that the proposed PPHS and MCHS strategies outperform the classic diagnosis-oriented item selection strategies in terms of bank utilization, and surpass the control-embedded strategies in terms of diagnostic performance. This pattern consistently emerges across different bank designs, bank-quality conditions, and AMP prior distributions. Our supplementary simulation further confirms that it remains unchanged when the bank size increases up to 600 (Table S19). These findings support our conclusion: the proposed strategies can conduct item selection in a way that enhances bank utilization at a smaller cost of diagnostic performance. Further, although the diagnostic accuracy of PPHS and MCHS is inferior to the diagnosis-oriented strategies in a traditional scenario (scenario 1), they may achieve comparable or even better results when the influence of item familiarity caused by item exposure is considered (scenario 2). This finding not only confirms the advantage of PPHS and MCHS over the classic control-embedded strategies but also implies their competitiveness in diagnostic accuracy in more realistic environments. The thorough inspection of bank utilization reveals another merit of PPHS and MCHS. That is, unlike DBS and GDI that show preference for items measuring one or a few attributes (Kaplan et al., 2015), PPHS and MCHS introduce no bias towards specific items. This merit avoids the security threat in the case that the given bank does not have enough items with the preferred pattern(s), which enhances the robustness of PPHS and MCHS under different bank settings. For the reason behind the advantage of the proposed strategies, a speculation is that instead of directly incorporating randomness into item selection, our path-searching idea enhances selection diversity by considering multiple items ahead, forming a variety of item combinations, and returning the first item in the combination that may best diagnose the current examinee. With the focus staying on accurate diagnosis, the negative impact of selection variety on the diagnostic performance could thus be minimized.

A notable cost of the proposed strategies is in their time complexity. PPHS and MCHS require longer time than the classic strategies because the tree-based search mechanism

Table 8 Results of the SHE variants on Bank A

Bank-Quality Condition	Indicator (mean+SD)	SHE	PPHS	MCHS	PPSHE	MCSHE
LD-LV	A1	**0.060 ± 0.013†**	0.086 ± 0.010*†	0.085 ± 0.017*†	**0.060 ± 0.013†**	0.072 ± 0.011*
	A2	**0.065 ± 0.010†**	0.093 ± 0.007*†	0.089 ± 0.009*†	0.067 ± 0.008†	0.076 ± 0.010*
	A3	**0.049 ± 0.007†**	0.079 ± 0.011*†	0.074 ± 0.011*†	0.052 ± 0.006†	0.057 ± 0.008*
	A4	0.048 ± 0.008†	0.081 ± 0.008*†	0.076 ± 0.010*†	**0.047 ± 0.006†**	0.055 ± 0.011*
	Pattern	**0.190 ± 0.017†**	0.281 ± 0.014*†	0.272 ± 0.018*†	0.194 ± 0.014†	0.220 ± 0.021*
	χ^2	68.027 ± 0.747†	16.460 ± 0.360*†	**14.209 ± 0.428*†**	67.843 ± 0.603†	49.037 ± 0.398*
	Overused	0.285 ± 0.002†	**0.226 ± 0.010*†**	0.265 ± 0.008*†	0.286 ± 0.004†	0.297 ± 0.006*
	Underused	0.446 ± 0.009†	**0.000 ± 0.000*†**	**0.000 ± 0.000*†**	0.448 ± 0.011†	0.327 ± 0.008*
LD-HV	A1	**0.057 ± 0.009**	0.080 ± 0.008*†	0.081 ± 0.013*†	**0.057 ± 0.009**	0.063 ± 0.011
	A2	**0.054 ± 0.007†**	0.091 ± 0.011*†	0.094 ± 0.009*†	0.056 ± 0.010†	0.066 ± 0.010*
	A3	**0.043 ± 0.007†**	0.070 ± 0.007*†	0.072 ± 0.011*†	0.046 ± 0.007†	0.054 ± 0.011*
	A4	**0.038 ± 0.007†**	0.067 ± 0.007*†	0.068 ± 0.008*†	0.040 ± 0.008	0.047 ± 0.008
	Pattern	**0.166 ± 0.010†**	0.261 ± 0.013*†	0.261 ± 0.018*†	0.169 ± 0.013†	0.198 ± 0.018*
	χ^2	70.732 ± 0.614†	19.796 ± 0.470*†	**17.792 ± 0.322*†**	70.503 ± 0.965†	51.549 ± 0.596*
	Overused	0.272 ± 0.005†	**0.224 ± 0.008*†**	0.250 ± 0.011*†	0.275 ± 0.004	0.278 ± 0.005
	Underused	0.455 ± 0.006†	**0.000 ± 0.000*†**	**0.000 ± 0.000*†**	0.458 ± 0.009†	0.307 ± 0.009*
HD-LV	A1	**0.006 ± 0.003**	0.017 ± 0.005*†	0.017 ± 0.006*†	**0.006 ± 0.002†**	0.008 ± 0.003*
	A2	**0.006 ± 0.003**	0.016 ± 0.005*†	0.014 ± 0.006*†	**0.006 ± 0.004**	0.008 ± 0.003
	A3	0.004 ± 0.002	0.012 ± 0.004*†	0.012 ± 0.003*†	**0.003 ± 0.002**	0.005 ± 0.002
	A4	0.004 ± 0.003†	0.010 ± 0.004*†	0.011 ± 0.004*†	**0.003 ± 0.003†**	0.007 ± 0.003*
	Pattern	0.019 ± 0.004†	0.051 ± 0.011*†	0.049 ± 0.010*†	**0.017 ± 0.004†**	0.026 ± 0.005*
	χ^2	49.697 ± 0.612*†	14.591 ± 0.378*†	**10.173 ± 0.239*†**	48.535 ± 0.442†	29.647 ± 0.239*
	Overused	0.297 ± 0.004†	**0.194 ± 0.006*†**	0.258 ± 0.008*†	0.301 ± 0.006†	0.291 ± 0.007*
	Underused	0.296 ± 0.009†	**0.000 ± 0.000*†**	**0.000 ± 0.000*†**	0.291 ± 0.011†	0.109 ± 0.009*
HD-HV	A1	0.005 ± 0.002†	0.013 ± 0.005*†	0.012 ± 0.002*†	**0.004 ± 0.003†**	0.007 ± 0.003*
	A2	**0.004 ± 0.002†**	0.013 ± 0.005*†	0.014 ± 0.004*†	**0.004 ± 0.002†**	0.007 ± 0.003*
	A3	**0.003 ± 0.002**	0.010 ± 0.004*†	0.009 ± 0.004*†	**0.003 ± 0.002**	0.004 ± 0.002
	A4	**0.002 ± 0.002†**	0.007 ± 0.004*	0.009 ± 0.004*†	**0.002 ± 0.002†**	0.005 ± 0.003*
	Pattern	**0.013 ± 0.004†**	0.040 ± 0.006*†	0.041 ± 0.005*†	**0.013 ± 0.005†**	0.021 ± 0.005*
	χ^2	53.997 ± 0.430*†	14.541 ± 0.284*†	**10.737 ± 0.296*†**	52.851 ± 0.381†	32.308 ± 0.414*
	Overused	0.290 ± 0.005†	**0.189 ± 0.008*†**	0.227 ± 0.010*†	0.287 ± 0.003†	0.278 ± 0.010*
	Underused	0.314 ± 0.011†	**0.000 ± 0.000*†**	**0.000 ± 0.000*†**	0.312 ± 0.007†	0.097 ± 0.011*

LD/HD: low/high item discrimination; LV/HV: low/high item variability; A*k*: attribute-level diagnostic error rate on attribute *k*; Pattern: pattern-level diagnostic error rate; χ^2: variance of item exposure rate; Overused/Underused: rates of overused/underused items; '*'/'†': significant difference with PPHS/MCHS at the 95% confidence level according to the Wilcoxon rank-sum test; **bold** number: best results among all strategies

is computationally intense. More specifically, given the bank size (I), attribute number (K), search depth (H), and search width (S), the time complexity of the fastest classic strategy SHE is $CC_{\text{SHE}} = O(I \cdot 2^K \cdot K)$, while that of PPHS and MCHS is $CC_{\text{PPHS}} = O(H \cdot I \cdot 2^{H+K} \cdot K)$ and $CC_{\text{MCHS}} = O(H \cdot S \cdot 2^{H+K} \cdot K)$. Given that the search width is always smaller than the bank size, the scale-up factors from CC_{SHE} to CC_{PPHS} and CC_{MCHS} are limited to $H \cdot 2^H$, demonstrating an exponential relationship with the search depth. The parameter investigation reveals that setting $H = 2$ holds promise and increasing H does not necessarily lead to better performance. Therefore, it can be considered that PPHS and MCHS can achieve promising results by increasing the time complexity by no more than eight times of SHE. This coincides with our simulation results, which show that the time cost of PPHS and MCHS is approximately 8–10 times and 1–4 times greater than SHE, respectively. Further, our simulation results show that even on the bank that measures five attributes with 600 items, PPHS and MCHS can select an item within 1 second (Fig. S7), which should go well beyond the perception of test takers. In summary, considering the short absolute time and the fast development

Table 9 Results of the SHE variants on Bank B

Bank B	Indicator (mean+SD)	SHE	PPHS	MCHS	PPSHE	MCSHE
LD-LV	A1	0.057 ± 0.009†	0.088 ± 0.010*†	0.086 ± 0.009*†	**0.054 ± 0.008†**	0.066 ± 0.009*
	A2	**0.049 ± 0.007†**	0.086 ± 0.010*†	0.081 ± 0.008*†	0.052 ± 0.006†	0.059 ± 0.009*
	A3	0.044 ± 0.007†	0.072 ± 0.010*†	0.075 ± 0.010*†	**0.042 ± 0.009†**	0.053 ± 0.009*
	A4	0.041 ± 0.006†	0.079 ± 0.009*†	0.077 ± 0.008*†	**0.038 ± 0.008†**	0.051 ± 0.005*
	A5	**0.045 ± 0.007†**	0.079 ± 0.010*†	0.079 ± 0.010*†	**0.045 ± 0.009†**	0.053 ± 0.005*
	Pattern	0.197 ± 0.009†	0.324 ± 0.016*†	0.318 ± 0.013*†	**0.194 ± 0.017†**	0.230 ± 0.016*
	χ^2	96.908 ± 0.648†	24.805 ± 0.558*†	**19.473 ± 0.561*†**	96.688 ± 0.608†	68.974 ± 0.541*
	Overused	0.251 ± 0.004†	**0.170 ± 0.009*†**	0.215 ± 0.008*†	0.251 ± 0.005†	0.257 ± 0.006*
	Underused	0.489 ± 0.006†	**0.000 ± 0.001*†**	**0.000 ± 0.001*†**	0.489 ± 0.005†	0.380 ± 0.010*
LD-HV	A1	0.053 ± 0.007†	0.087 ± 0.011*†	0.090 ± 0.010*†	**0.051 ± 0.007†**	0.066 ± 0.011*
	A2	**0.043 ± 0.007†**	0.081 ± 0.010*†	0.077 ± 0.006*†	0.044 ± 0.008†	0.056 ± 0.008*
	A3	0.037 ± 0.005†	0.067 ± 0.010*†	0.064 ± 0.009*†	**0.036 ± 0.005†**	0.047 ± 0.007*
	A4	0.035 ± 0.007†	0.069 ± 0.009*†	0.068 ± 0.009*†	**0.034 ± 0.006†**	0.045 ± 0.009*
	A5	0.035 ± 0.006†	0.068 ± 0.007*†	0.064 ± 0.010*†	**0.033 ± 0.007†**	0.041 ± 0.007*
	Pattern	0.173 ± 0.009†	0.298 ± 0.013*†	0.293 ± 0.012*†	**0.170 ± 0.009†**	0.213 ± 0.020*
	χ^2	105.491 ± 0.603†	26.823 ± 0.568*†	**22.195 ± 0.368*†**	105.083 ± 0.794†	74.344 ± 0.658*
	Overused	0.235 ± 0.004†	**0.176 ± 0.004*†**	0.193 ± 0.004*†	0.232 ± 0.004	0.231 ± 0.004
	Underused	0.493 ± 0.005†	0.001 ± 0.003*†	**0.000 ± 0.001*†**	0.492 ± 0.005†	0.354 ± 0.008*
HD-LV	A1	**0.003 ± 0.002†**	0.014 ± 0.004*†	0.014 ± 0.004*†	**0.003 ± 0.002†**	0.005 ± 0.002*
	A2	0.003 ± 0.002	0.012 ± 0.004*†	0.010 ± 0.005*†	**0.002 ± 0.001†**	0.003 ± 0.002*
	A3	**0.002 ± 0.002**	0.011 ± 0.004*†	0.009 ± 0.003*†	**0.002 ± 0.001**	0.004 ± 0.002
	A4	0.002 ± 0.002	0.010 ± 0.003*†	0.009 ± 0.003*†	**0.001 ± 0.001†**	0.003 ± 0.002*
	A5	**0.002 ± 0.002†**	0.011 ± 0.003*†	0.009 ± 0.003*†	**0.002 ± 0.002†**	0.004 ± 0.002*
	Pattern	0.012 ± 0.004†	0.055 ± 0.007*†	0.048 ± 0.008*†	**0.010 ± 0.003†**	0.018 ± 0.004*
	χ^2	77.102 ± 0.601*†	22.619 ± 0.527*†	**14.927 ± 0.338*†**	76.250 ± 0.598†	45.602 ± 0.408*
	Overused	0.246 ± 0.004	**0.154 ± 0.007*†**	0.183 ± 0.006*†	0.247 ± 0.003	0.248 ± 0.005
	Underused	0.373 ± 0.006†	**0.000 ± 0.001*†**	**0.000 ± 0.001*†**	0.370 ± 0.007†	0.176 ± 0.013*
HD-HV	A1	0.003 ± 0.002	0.014 ± 0.004*†	0.011 ± 0.004*†	**0.002 ± 0.002†**	0.004 ± 0.002*
	A2	**0.001 ± 0.002†**	0.011 ± 0.004*†	0.010 ± 0.003*†	0.002 ± 0.002†	0.003 ± 0.002*
	A3	**0.001 ± 0.001†**	0.009 ± 0.004*†	0.007 ± 0.002*†	**0.001 ± 0.002†**	0.003 ± 0.002*
	A4	**0.001 ± 0.002†**	0.008 ± 0.003*†	0.007 ± 0.003*†	**0.001 ± 0.002†**	0.003 ± 0.002*
	A5	**0.001 ± 0.002†**	0.008 ± 0.004*†	0.006 ± 0.002*†	**0.001 ± 0.001†**	0.003 ± 0.002*
	Pattern	0.008 ± 0.004†	0.048 ± 0.008*†	0.039 ± 0.006*†	**0.007 ± 0.003†**	0.015 ± 0.003*
	χ^2	83.505 ± 0.499*†	23.255 ± 0.340*†	**15.705 ± 0.331*†**	82.061 ± 0.432†	50.410 ± 0.486*
	Overused	0.237 ± 0.006†	**0.129 ± 0.005*†**	0.159 ± 0.007*†	0.237 ± 0.005†	0.224 ± 0.006*
	Underused	0.380 ± 0.007†	**0.000 ± 0.001*†**	**0.000 ± 0.000*†**	0.375 ± 0.006†	0.169 ± 0.012*

LD/HD: low/high item discrimination; LV/HV: low/high item variability; A*k*: attribute-level diagnostic error rate on attribute *k*; Pattern: pattern-level diagnostic error rate; χ^2: variance of item exposure rate; Overused/Underused: rates of overused/underused items; '*'/'†': significant difference with PPHS/MCHS at the 95% confidence level according to the Wilcoxon rank-sum test; **bold** number: best results among all strategies

of computers, the computational time should not be considered as the bottleneck that constrains the practical use of the proposed strategies.

Recommendation on item selection strategies

The simulation results show that the two proposed strategies and the four classic item selection strategies (i.e., SHE, GDI, RP_PWKL, and DBS) all have their own merits and deficiencies. To prepare for practical use, we make the following recommendation.

The diagnosis-oriented strategies such as SHE and GDI generally achieve the best diagnostic accuracy and efficiency, but also have the most imbalanced bank utilization. They are thus suitable for the scenarios that emphasize diagnostic performance but pay little attention to item exposure control

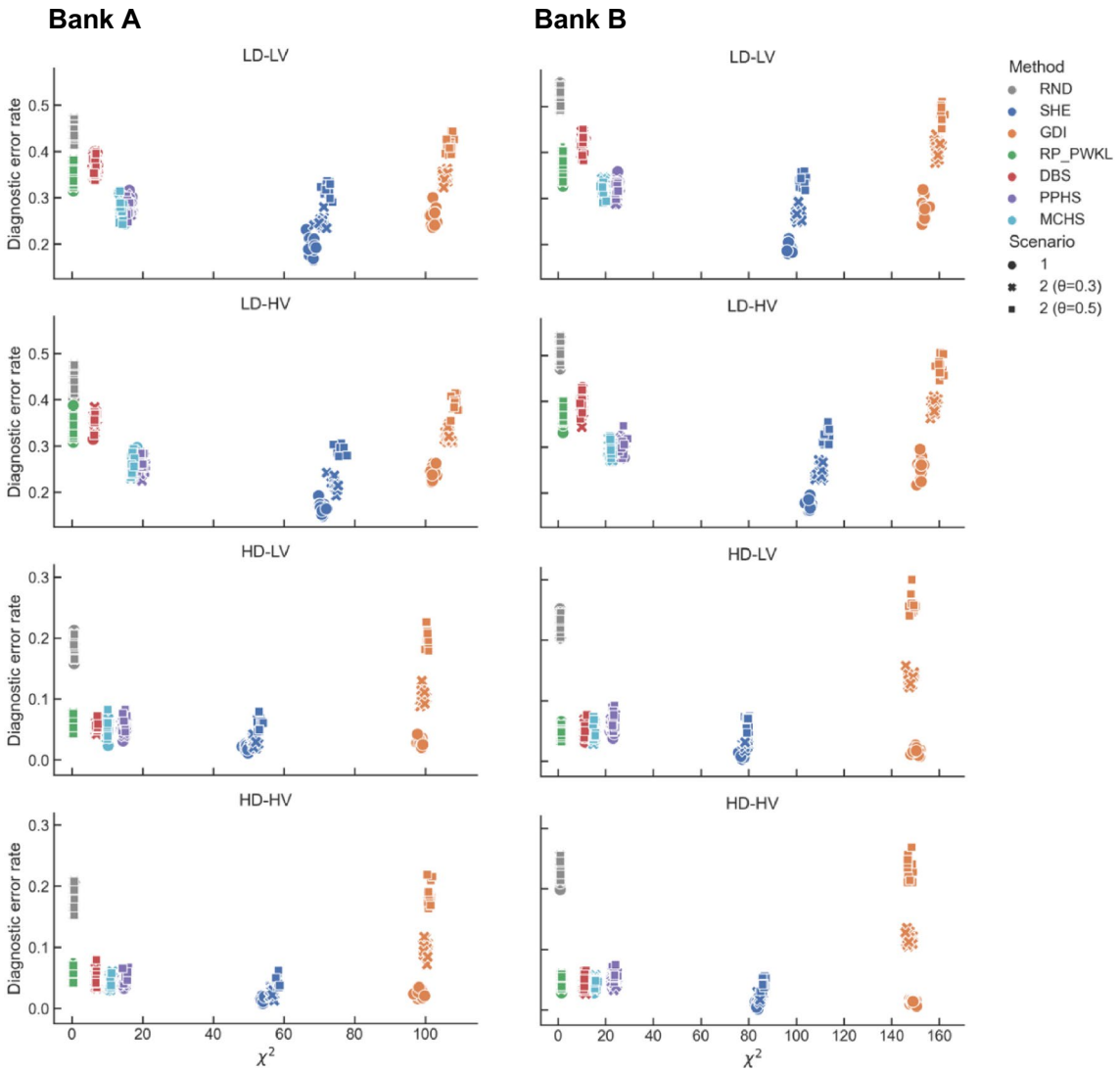

Fig. 7 Comparison on the overall performance of item selection strategies across scenarios

(e.g., low-stakes assessments). Our simulation results further reveal that GDI is inferior to SHE across all evaluation criteria except for the item overused rate, which suggests that GDI may be a more appropriate choice between these two strategies if test security must be considered.

The control-embedded strategies such as RP_PWKL and DBS have the best bank utilization, yet their diagnostic performance is often the worst. They are thus more appropriate for the scenarios that favor balanced bank utilization. Besides, our simulations indicate that RP_PWKL and DBS can achieve satisfying accuracy given that the bank quality is high, which suggests that they may be a good fit for high-stakes tests that have access to high-quality items. Note that both of these strategies have excellent control over item underutilization, keeping the underused rate close to zero across various conditions. However, only RP_PWKL achieves excellent control over item overexposure. Therefore, RP_PWKL is particularly recommended for the scenarios where security threats must be minimized.

The two proposed strategies, PPHS and MCHS, consistently outperform the diagnosis-oriented strategies in terms of bank utilization while maintaining solid advantage over the control-embedded strategies in terms of diagnostic performance. In particular, PPHS and MCHS also have excellent control over item underutilization, and they both outperform DBS in most case for suppressing item overexposure. As a result, the proposed strategies are suitable for the scenarios that need to balance test accuracy and security while minimizing the waste of bank capacity. Further, PPHS achieves a lower item overexposure rate but exhibits a higher item exposure variance than MCHS, which suggests that PPHS may have fewer overexposed items with a

higher overexposure degree. The decision to choose between PPHS and MCHS should therefore be made with caution. MCHS may exhibit better performance if a higher threshold is allowed for overexposure. In addition, based on the simulation results in scenario 2, the difference in the diagnostic performance between the diagnosis-oriented strategies and the proposed strategies may be further reduced due to the familiarity effect that exists in the real world. Therefore, given sufficient computational resources, we would encourage users to test the proposed strategies first before choosing the others.

As for the two parameters of the proposed strategies (i.e., search depth and width), we have the following suggestions. Firstly, for both PPHS and MCHS, the search depth of two is a promising setting that can strike a satisfying balance between diagnostic performance and bank utilization with a reasonable time cost. According to the results of our parameter investigation, increasing the search depth may not bring further improvement in diagnostic accuracy because the estimation error would accumulate as the search branch grows in depth. It may also worsen the bank security and bring super-linear growth in computational time. Secondly, for PPHS, the search width of three is a generally good choice for balancing algorithmic performance and computational time. Since the increment in computational time is mostly linear as the search widens, users may further increase the search width for better accuracy on large item banks, but be cautious, as it may impair bank utilization. Third, MCHS is more sensitive to the setting of search width. We consider 200/300 to be a good initial setting and users may consider fine-tuning it for large item banks through ad hoc simulations.

Limitations and future directions

Several limitations need further consideration. Firstly, a general assumption of the current study is that parameters of items in the banks are estimated without errors. However, in practice, the estimation of item parameters is often biased (Li & Suen, 2013; Lim & Drasgow, 2017). It has been reported that these errors can affect the performance of item selection criteria and strategies, particularly when the sample size is small (Huang, 2018; Şahin & Weiss, 2015; van der Linden & Glas, 2000b). Therefore, future studies should conduct further investigations to examine the performance of the proposed PPHS and MCHS strategies when different types of estimation errors are presented. In addition, the relationship among attributes may be more complex in the real world than in our simulation studies (Kuo et al., 2016; Ma et al., 2023). Future research should improve the current design by taking interdependency and hierarchy among attributes into account. As for practical constraint, the current study only applies the item type constraint and the test length constraint. Future studies may explore more practical constraint settings like the item type constraint on tests of variable length. Additionally, we only verify our algorithm through simulations with virtual examinees because the true AMPs of virtual examinees are known and so far there have not been publicly available item banks suitable for our study. It is vital that future studies apply the proposed strategies in practice to verify their effectiveness.

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Authors' contributions X. Cao and Y. Lin conceived the present idea, designed the algorithms, and organized the experiments. X. Cao implemented the algorithms, performed the experiments, and analyzed the results. All authors participated in result discussions and contributed to the writing and/or revision of the manuscript.

Declarations

Consent to publication All the authors gave their consent to the publication of this work.

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